

Theme: Sustainable city

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UNIT 01

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By the end of the theme, students will be able to:

- Recognize energy transformation processes in wind turbines.
- Identify different types of sensors and their practical applications.
- Comprehend the working of RFID technology, including data reading from RFID tags and transmission to a receiver.
- Understand the environmental impact of various energy sources, highlighting the advantages of renewable energy, particularly wind power.
- Understand the concept of automation and its advantages.
- Understand energy transformation processes in wind turbines. (Note: This is repeated in the book)
- Strengthen hands-on skills in building and testing electrical circuits.
- Learn how to control home devices using a mobile application.
- Program an Arduino to process RFID data and control a garage door mechanism.
- Analyze the efficiency of energy conversion in wind turbines, considering factors such as blade design and wind speed.
- Explore the environmental impact of various energy sources, highlighting the advantages of renewable energy, particularly wind power. (Note: This is repeated in the book with slightly different wording)
- Incorporate water conservation and waste management systems that complement energy systems in city designs.
- Design urban green spaces and forests to enhance air quality and provide recreational areas.
- Develop a city model that integrates renewable energy sources for efficient energy use.
- Design and build a basic circuit that integrates an RFID reader and operates a servo motor development.

Objectives:

- 1-Understand the core challenges facing urban environments in terms of energy demands and environmental concerns.
- 2-Identify the central problem presented in the "Sustainable City" project scenario involving Adam and Laila.
- 3-Recognize that the Engineering Design Process (EDP) will be the framework used to address the project task.
- 4-Begin to connect the broad learning objectives of the unit (as outlined on Page 2) to the specific challenges of the case scenario.
- 5-Express initial curiosity and engagement with the theme of sustainable cities and potential solutions.

Challenge Questions of the Lesson

- 1-Why is the concept of a "sustainable city" becoming increasingly important in today's world?
- 2-How can understanding the specific challenges faced by Adam and Laila's city help us think about broader issues of urban sustainability?
- 3-What role do you think the Engineering Design Process will play in helping Adam and Laila (and us) find solutions?
- 4-Based on the initial scenario and the learning objectives, what are some of the key areas we will likely need to explore in this unit?

here are short and simple answers to the Challenge Questions :

- **Why sustainable cities matter:** Cities use lots of energy and can harm the Earth, so being sustainable helps everyone now and later.
- **Adam & Laila's city helps us:** Their problems show us the big issues cities face with energy and the environment.
- **Role of EDP:** The EDP is like a plan to help us solve their problems step by step.
- **Key areas to explore:** We'll likely learn about new energy, how things work, and how to build better cities.

SEL relation

- **Self-Awareness:** Students begin to connect their own understanding of environmental issues and energy use to the presented scenario.
- **Relationship Skills:** Students engage in discussions and share their initial thoughts and ideas about the project.
- **Responsible Decision-Making:** Students start to understand the importance of making informed decisions about energy and environmental sustainability.
- **Curiosity:** Students are encouraged to ask questions and express their interest in the topic of sustainable cities and the project challenge.

Introducing the Sustainable City Challenge

Teacher Role:

- Introduce the theme of "Sustainable City" and its relevance to current global challenges related to energy and the environment.
- Present the case scenario involving Adam and Laila and their city's increasing energy demands and environmental concerns.
- Highlight the core problem that students will be addressing: investigating alternative energy sources for sustainability.
- Engage students by discussing their own understanding of "sustainability" and examples of environmental concerns in their local or global context.
- Set the stage for the project by emphasizing that they will be helping Adam and Laila find solutions.

Expected From Students:

- Begin to understand the concept of a "Sustainable City" and its importance.
- Identify the core challenges presented in the case scenario (increasing energy demands and environmental concerns).
- Recognize the role of Adam and Laila as relatable characters facing a problem they need to solve.
- Start thinking about different types of "alternative energy sources" and their potential benefits for a city.
- Express their initial thoughts and questions about sustainability and potential solutions.

Strategies to use with this part:

1. Preview with a Hook

- **Strategy:** Begin with a visual or question that connects to the theme of a sustainable city.
- **Implementation:**
 - Display contrasting images: one of a polluted, resource-depleted city versus one of a clean, green city with renewable energy sources.
 - Ask: "Which of these cities looks more appealing to live in? What are the key differences you notice?"
 - Briefly discuss students' initial ideas about what makes a city sustainable.

2. Make Predictions

- **Strategy:** Use the theme title and the initial scenario to spark curiosity about the project.
- **Implementation:**
 - Show the theme title: "Sustainable City" and the lesson title: "Case Scenario."
 - Ask: "Based on the theme and the fact that Adam and Laila's city has energy and environmental problems, what kind of solutions do you think they will be looking for?"
 - Have students share their initial predictions about the project's focus.

3. Set the Scene with a Visual

- **Strategy:** Use the image on Page 1 to help students visualize the context.

- **Implementation:**

- Direct students to the image on Page 1 showing a cityscape with a mix of traditional and potentially sustainable elements (like wind turbines in the distance).
- Ask: "What do you notice in this picture of Adam and Laila's city? What are some signs of modern life, and what potential for cleaner energy do you see?"

4. Think-Pair-Share

- **Strategy:** Encourage students to reflect on the challenges presented in the overview.
- **Implementation:**
 - After reading the "Project Overview," ask: "What are the two main problems facing Adam and Laila's city?"
 - Have students think about their answer individually, then discuss it with a partner.
 - Invite a few pairs to share their understanding of the core challenges with the class.

5. Interactive Questioning

- **Strategy:** Engage students by having them step into the shoes of Adam and Laila.
- **Implementation:**

- Ask: "If you were Adam or Laila, living in a city with increasing energy problems and environmental worries, what would be the first thing you would want to investigate?"

- Have students share their initial ideas about where they would start their search for solutions.

6. Connecting to Prior Knowledge

- **Strategy:** Activate students' existing knowledge about energy and the environment.
- **Implementation:**
 - Ask: "What do you already know about different ways cities get their energy? What are some of the negative effects of how cities use energy now?"
 - Briefly discuss their existing understanding to bridge it with the new theme.

Materials Needed:

- Student books with Page 1 of Lesson 1.
- (Optional) Visual aids or real-world examples illustrating energy demands and environmental issues in urban areas.

Step-by-Step Instructions:

1. Introduce the Theme (5-10 minutes):

- Begin by introducing the overarching theme of "Sustainable City." Briefly discuss what makes a city sustainable (meeting the needs of the present without compromising the ability of future generations to meet their own needs).
- Ask students for examples of why sustainability is important in today's world.

2. Present the Case Scenario (10-15 minutes):

- Direct students to Page 1 and read the "Project Overview" aloud or have them read it silently.
- Emphasize the challenges faced by Adam and Laila's city.
- Ask students to relate these challenges to real-world situations they may be aware of (e.g., power outages, pollution, climate change).

3. Brainstorm Alternative Energy Sources (10-15 minutes):

- Initiate a class discussion on "alternative energy sources." Ask students what types of energy sources they know that are not fossil fuels (e.g., solar, wind, hydro).
- Briefly discuss the potential benefits of these sources in making a city more sustainable.

4. Setting the Stage for the Project (5 minutes):

- Explain to students that throughout this unit, they will be working to help Adam and Laila find and understand alternative energy solutions for their city.
- Highlight that this first lesson is about understanding the problem they need to solve.

Differentiation:

- **Support:** Provide students with visual aids or simplified explanations of energy demands and environmental concerns. Offer a list of alternative energy sources as a starting point for brainstorming.
 - **Extension:** Encourage students to research current energy challenges in specific cities or countries. They could also start thinking about the pros and cons of different alternative energy sources.
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Learning Objectives and Project Task

Teacher Role:

- Clearly present and explain the Learning Objectives for the theme, emphasizing what students will gain knowledge and skills in.
- Facilitate discussions to connect the Learning Objectives to the Case Scenario presented on Page 1, helping students see the relevance of the learning.
- Introduce the Project Task, breaking it down and clarifying the expectation of using the Engineering Design Process.
- Encourage student curiosity and questions about the objectives and the project.
- Set a positive and engaging tone for the unit, highlighting the exciting challenge of designing a sustainable city.

Expected From Students:

- Actively listen and engage with the explanation of the Learning Objectives.
- Begin to understand how the Learning Objectives relate to the challenges faced by Adam and Laila in the Case Scenario.
- Understand the core task of the project: to follow the Engineering Design Process to create a sustainable city plan.
- Ask clarifying questions about the Learning Objectives and the Project Task.
- Express initial enthusiasm and ideas about the project.

Strategies to use with this part:

1. Objective Preview (Think-Pair-Share)

- **Strategy:** Introduce the learning objectives by having students consider what they will be learning.
- **Implementation:**
 - Display a few of the learning objectives (e.g., "Recognize energy transformation processes in wind turbines," "Identify different types of sensors").
 - Ask: "What do you think you will be learning about in this theme based on these objectives?"
 - Have students think individually, pair up to discuss their interpretations, and then share with the class.

2. Connecting Objectives to the Scenario (Discussion)

- **Strategy:** Link the learning objectives back to the problem faced by Adam and Laila.
- **Implementation:**
 - For a few key objectives, ask: "How might understanding [specific objective, e.g., different types of sensors] help Adam and Laila in their quest for sustainable energy solutions?"
 - Facilitate a class discussion to brainstorm connections.

3. Visualizing the Project (Brainstorming)

- **Strategy:** Encourage students to imagine the final outcome of the project.
- **Implementation:**

- Ask: "What might a 'sustainable city' designed by you and your classmates look like? What kinds of energy sources and technologies might it include?"
- Have students brainstorm ideas individually or in pairs and share their visions.

4. **Setting Expectations (Teacher Explanation)**

- **Strategy:** Clearly explain the overall expectations and the role of the learning objectives in achieving the Project Task.
- **Implementation:**
 - Explicitly state that the learning objectives on the page are the building blocks that will equip them with the knowledge and skills needed to successfully complete the Project Task.
 - Briefly outline how the lessons in the unit will contribute to achieving these objectives.

Objectives:

- 1-Define the Engineering Design Process (EDP) and recognize its importance in problem-solving.
- 2-Identify and describe the five key stages of the EDP: Ask, Imagine, Plan, Create, and Improve.
- 3-Understand the cyclical and iterative nature of the EDP.
- 4-Begin to apply the EDP framework to the context of the Sustainable City project and the challenges faced by Adam and Laila.
- 5-Recognize that the EDP will guide their work throughout the unit.

Challenge Questions of the Lesson

- 1-Why is it helpful to have a structured process like the Engineering Design Process when trying to solve complex problems?
- 2-In your own words, what is the main goal of each stage of the EDP (Ask, Imagine, Plan, Create, Improve)?
- 3-How might the cyclical nature of the EDP be beneficial when working on a long-term project like designing a sustainable city?
- 4-How can the EDP help us to effectively address the needs and constraints identified in the "Ask" stage of the Sustainable City project?

SEL relation

- **Self-Awareness:** Students reflect on their own problem-solving approaches and begin to understand the value of a structured methodology.
- **Self-Management:** Students learn to organize their thinking and approach to the project using the EDP framework.
- **Collaboration:** Students may discuss the EDP stages in groups, fostering shared understanding.
- **Critical Thinking:** Students analyze the steps involved in the EDP and consider how they apply to the specific challenges of the project.
- **Growth Mindset:** Students understand that the "Improve" stage emphasizes learning from mistakes and iterating towards better solutions.

here are short and simple answers to the Challenge Questions :

- **Why a process helps:** It gives us clear steps to solve problems.
- **EDP stage goals:**
 - **Ask:** Find out what the problem is.
 - **Imagine:** Think of lots of ideas.
 - **Plan:** Choose the best ideas and how to do them.
 - **Create:** Make a model or try it out.
 - **Improve:** Make it better after testing.
- **Why the cycle is good:** We can try things, learn, and make our ideas better.
- **How EDP helps our project:** It helps us find good ways to solve Adam and Laila's city problems.

Engineering Design Process

Teacher Role:

- Introduce the Engineering Design Process (EDP) as a systematic approach to problem-solving that will be used throughout the Sustainable City project.
- Explain each stage of the EDP (Ask, Imagine, Plan, Create, Improve) in a clear and concise manner, relating it to the challenge presented in Lesson 1 (helping Adam and Laila).
- Emphasize the cyclical nature of the EDP and how the "Improve" stage can lead back to earlier stages for refinement.
- Guide students to understand that this lesson is about familiarizing themselves with the framework they will use for their project.
- Encourage students to start thinking about how the EDP can be applied to the problem of creating a more sustainable city for Adam and Laila.

Expected From Students:

- Understand that the Engineering Design Process (EDP) is a structured way to solve problems.
- Learn the names and a basic description of each stage of the EDP (Ask, Imagine, Plan, Create, Improve).
- Recognize the cyclical nature of the EDP and the importance of iteration and improvement.
- Begin to think about how the EDP stages might relate to the task of designing a sustainable city for Adam and Laila.

- Be prepared to use the EDP framework in subsequent lessons to develop their project.

Strategies to use with this part:

1. Visual Introduction (EDP Diagram)

- **Strategy:** Use the visual representation on the page as the central focus for introducing the EDP.
- **Implementation:** Point to each stage of the cycle as you explain it. Use gestures to emphasize the cyclical flow and the connections between the stages.

2. Real-World Examples (Brainstorming)

- **Strategy:** Connect the abstract concept of the EDP to familiar situations or simple problems.
- **Implementation:** Ask students: "Think about a time you tried to solve a problem, like building with LEGOs or fixing a toy. Can you see how you might have gone through some of these stages without even realizing it?" Guide them to identify parallels with the EDP stages.

3. "What Happens Next?" (Sequencing)

- **Strategy:** Reinforce the order of the EDP stages and the flow of the process.

- **Implementation:** Present a simple scenario related to sustainability (e.g., wanting to reduce waste in the classroom). Ask students: "What would be the first step in

using the EDP to solve this? What would come next after we brainstorm ideas?" Guide them through the sequence.

4. Role-Playing the EDP (Small Groups)

- **Strategy:** Engage students actively by having them role-play the different stages of the EDP in the context of the Sustainable City project.
- **Implementation:** Divide students into small groups, assign each group a stage of the EDP, and have them briefly discuss what actions they would take during that stage for Adam and Laila's city.

5. Think-Pair-Share (Applying to the Project)

- **Strategy:** Encourage individual reflection and peer discussion on how the EDP relates to the main project.
- **Implementation:** Ask: "Which stage of the EDP do you think will be most critical for successfully designing a sustainable city? Why?" Have students think individually, discuss with a partner, and then share their ideas with the class.

Step-by-Step Instructions:

1. Review the Project Challenge (5-10 minutes):

- Briefly remind students of the challenge presented in Lesson 1: helping Adam and Laila make their city more sustainable.

2. Introduce the Engineering Design Process (10-15 minutes):

- Present the EDP as a tool that engineers and designers use to solve problems.
- Explain that they will be using this process to guide their work on the Sustainable City project.

- Go through each stage of the EDP diagram on Page 1:
 - **Ask:** Understanding the problem, identifying needs and constraints. (Relate to Adam and Laila's energy and environmental issues).
 - **Imagine:** Brainstorming possible solutions and ideas. (Think about different alternative energy sources).
 - **Plan:** Selecting the best ideas and developing a detailed plan. (How will they integrate these solutions into a city?).
 - **Create:** Building a prototype or model (This might be a conceptual model or a physical one later).
 - **Improve:** Testing and evaluating the solution, then making changes for better results. (How can their sustainable city design be made more effective?).
- Emphasize the arrows showing that the process is often iterative, and you might go back to earlier stages.

3. Applying EDP to the Project (15-20 minutes):

- Lead a class discussion on how the EDP might apply to the Sustainable City project.
- Ask questions like:
 - "In the 'Ask' stage, what questions do we need to answer about Adam and Laila's city?"

- "In the 'Imagine' stage, what are some initial ideas for making their city more sustainable?"
- "What might be involved in the 'Plan' stage of this project?"

- "What could a 'Create' stage look like for our project (initially)?"
- "How might we 'Improve' our sustainable city design?"

4. Individual Reflection (5 minutes):

- Have students individually think about which stage of the EDP seems most important or challenging to them for this project and why.

Differentiation:

- **Support:** Provide students with a worksheet that defines each stage of the EDP with simple language and guiding questions related to the Sustainable City project.
- **Extension:** Challenge students to research real-world examples of sustainable city initiatives and identify which stages of the EDP were likely involved in their development.

Engineering Design Process

LESSON 01

Name: _____

Project: _____

Ask



Imagine

Improve

Plan

Create



Chapter 1: Design

Objectives:

- 1-Describe key characteristics and features of a smart, eco-friendly city.
 - 2-Identify and explain various components that contribute to the sustainability of a city.
 - 3-Analyze real-world examples of sustainable city initiatives in Copenhagen, Amsterdam, and Singapore.
 - 4-Understand the importance of clean energy and be able to communicate its benefits through a visual medium (poster).
- Critically evaluate how geographical location and climate influence the selection and implementation of sustainable energy solutions in a given city.

Challenge Questions of the Lesson

- 1-What are the most significant differences between a conventionally developed city and a truly sustainable city?
- 2-How can learning from the successes and challenges of cities like Copenhagen, Amsterdam, and Singapore inform our own efforts to build sustainable communities?
- 3-Why is the integration of diverse components (like transportation, energy, waste management, and green spaces) crucial for achieving urban sustainability?
- 4-How should the unique geographical and climatic conditions of a city influence its strategies for adopting sustainable energy and other eco-friendly practices?

SEL relation

* **Self-Awareness:** Students reflect on their own perceptions of their city and compare it to the concept of a sustainable city.

* **Social Awareness:** Students develop a broader understanding of global efforts towards urban sustainability by learning about different cities' initiatives.

* **Critical Thinking:** Students analyze and evaluate the effectiveness of different sustainable practices and consider the influence of various factors.

* **Creativity:** Students express their understanding of clean energy through the creation of a poster.

* **Decision-Making:** Students consider the factors that influence the choice of sustainable energy solutions for a specific context.

here are short and simple answers to the Challenge Questions:

- **City differences:** Sustainable cities are cleaner, greener, and use resources wisely.
- **Learning from other cities:** Their ideas can help us make our cities better too.
- **Why mix components:** Everything in a city is connected, so we need to improve it all together.
- **Geography and climate matter:** What works in one city might not work in another because of weather and location.

1.Let's think & Explore

Teacher Role:

- Initiate a class discussion using the "Let's Think" questions to encourage students to envision and compare their current city with an ideal sustainable city.
- Introduce the "Explore" activity as a virtual field trip to showcase real-world examples of successful sustainable city initiatives.
- Highlight the focus areas for the virtual field trip: transportation, waste management, and urban planning.
- Encourage active note-taking during the virtual exploration, prompting students to consider the applicability of these ideas to their own community or the project city of Adam and Laila.
- Point out the "Watch it.." and "Read about it" resources as supplementary materials for further learning and inspiration.

Expected From Students:

- Engage in thoughtful reflection and discussion about the characteristics of a smart, eco-friendly city.
- Compare and contrast their current urban environment with their imagined sustainable city, considering the provided aspects.
- Understand the purpose of the virtual field trip: to learn from the sustainability strategies of Copenhagen, Amsterdam, and Singapore.
- Actively participate in the virtual exploration (if conducted in class) or independently research these cities.

- Take notes on the specific initiatives observed in the virtual field trip, focusing on transportation, waste management, and urban planning.
- Begin to think critically about how the learned strategies could be adapted and applied to their own community or Adam and Laila's city.
- Utilize the provided "Watch it.." and "Read about it" resources for deeper understanding.

Materials Needed:

- Student books with Page 1 of Lesson 3.
- Internet access and devices to conduct the virtual field trip (if done in class).
- Whiteboard or projector to facilitate discussion and note-taking.
- (Optional) Access to the videos and articles linked via the QR codes or website addresses.

Step-by-Step Instructions:

1. Engage with "Let's Think" (15-20 minutes):

- Begin by asking students to close their eyes and imagine living in a truly smart and eco-friendly city. Use the first "Let's Think" question as a starting point.
- Facilitate a class discussion where students share their visions. Write down key features they mention on the board.
- Move on to the second and third "Let's Think" questions, prompting students to compare their imagined city with their current one, focusing on energy, transportation, waste, and the environment. Encourage them to think about specific improvements.

2. Introduce "Explore - Do it" (10-15 minutes):

- Introduce the virtual field trip to Copenhagen, Amsterdam, and Singapore as examples of cities with notable sustainability efforts.
- Explain that they will be exploring these cities through online resources, videos, or pre-selected information.
- Highlight the three key areas to focus on during their exploration: transportation, waste management, and urban planning.
- Emphasize the importance of taking notes on specific strategies and initiatives they observe.

3. Conduct the Virtual Field Trip (Time Varies):

- This can be done as a guided in-class activity using online resources and visuals, or as an independent research task.
- Provide students with starting points for their research on each city (e.g., specific websites, videos).
- Encourage them to look for concrete examples of sustainable practices in the three focus areas.

4. Discuss Findings and Application (15-20 minutes - after the virtual trip):

- After the virtual exploration, facilitate a class discussion where students share their key findings for each city.
- Focus the discussion on the last prompt in the "Explore" section: "consider how we can apply their ideas to make our own community more eco-friendly." Encourage students to think about their local context or Adam and Laila's city.

5. Introduce "Watch it.." and "Read about it" (5 minutes):

- Point out the QR codes and website links as additional resources they can use for further learning about sustainable cities.
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2.Focus: Identifying Key Components of a Sustainable City

Teacher Role:

- Guide students to analyze the provided image of a sustainable city and identify its key components.
- Facilitate a discussion to ensure students understand the meaning of each labeled component in the image.
- Introduce the table as a tool for students to consolidate their understanding of these components by writing concise descriptions.
- Encourage students to use information gathered during the virtual field trip (from Lesson 3, Page 1) and their prior knowledge to complete the table.
- Review and discuss the completed table, providing the correct answers to reinforce the key elements of a sustainable city.

Expected From Students:

- Carefully examine the image of the sustainable city and identify the labeled components.
- Understand the general function and purpose of each component within the context of sustainability (e.g., renewable energy, green space, sustainable transportation).

- Use their understanding and any notes from the virtual field trip to write brief descriptions for each component in the table.
- Participate in discussions about the components and their role in creating a sustainable urban environment.
- Demonstrate an understanding of the interconnectedness of different systems within a sustainable city by comparing their answers to the provided correct descriptions.

Materials Needed:

- Student books with Page 2 of Lesson 3.
- Whiteboard or projector to display the image and table for class discussion.
- Markers or pens for students to fill in their tables.
- **Answer Key for the Table:**

Component	Description
Intelligent energy distribution systems	Intelligent energy distribution systems that optimize efficiency and reliability.
Sustainable Transportation	Eco-friendly modes of transport, such as public transit, cycling, and walking.
Employment Opportunities	Creation of jobs related to sustainability, such as renewable energy and green technology.
Community owned infrastructure	Shared ownership and management of public facilities to promote community engagement
Resource-efficient Buildings	Structures designed to minimize energy and water consumption.
Remanufacturing and Recycling	Processes that reuse materials and reduce waste through repair, refurbishment, and material recovery.
Retrofitting Existing Buildings	Modernizing older buildings to improve energy efficiency and sustainability.
Renewable Energy at Community Level	Generation of clean energy sources like solar, wind, or hydro within local communities.

Waste-to-Energy Plants	produces energy thanks to the heat generated by the combustion of municipal and industrial solid waste
Green Space	Parks, gardens, and other natural areas that provide environmental and social benefits.
Waste-to-Energy	Conversion of waste into energy through various processes.
Combined Heat and Power plants	Facilities that produce both electricity and heat, improving energy efficiency.
Water efficiency	Implementing technologies and practices to reduce water consumption and loss.

Step-by-Step Instructions:

1. Review the Virtual Field Trip (10-15 minutes):

- Begin by briefly revisiting the cities explored in the virtual field trip (Copenhagen, Amsterdam, Singapore).
- Ask students to recall specific examples of sustainable transportation, waste management, and urban planning they observed in those cities.
- Connect these real-world examples to the idea of a comprehensive "sustainable city" model.

2. Analyze the Sustainable City Image (15-20 minutes):

- Direct students to the image on Page 2.
- Go through each labeled component in the image, asking students what they understand about its role in a sustainable city. Facilitate a brief discussion for each component to ensure clarity, including the concept of community involvement in public resources.

3. Complete the Table (15-20 minutes):

- Explain the task: students need to fill in the "Description" column of the table for the provided "Component" names, ensuring they include the newly identified component.
- Encourage them to use their understanding from the image analysis and any relevant notes from the virtual field trip.
- Circulate and provide support as students work on the table.

4. Class Discussion and Review with Answer Key (10-15 minutes):

- Once students have had time to complete the table, review the descriptions as a class.
- Ask students to share their descriptions.
- **Display the Revised Answer Key (either verbally or visually) and discuss the correct descriptions for each component, including the "Shared ownership and management of public facilities."**
- Address any misconceptions or variations in student answers.
- Reinforce the key characteristics and benefits of each component in creating a sustainable city.

Differentiation:

- **Extension:** Challenge students to research one specific component in more detail and present its benefits and implementation strategies to the class. They could also think about components that are not explicitly listed but are important for sustainability and community involvement.
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3.Practice & Challenge

Teacher Role:

- Guide students to review the table provided, ensuring they understand the components and their descriptions related to a sustainable city.
- Introduce the "Practice" activity: creating a poster about clean energy. Explain the purpose of the poster in raising awareness and understanding.
- Encourage creativity and the integration of information learned during the lesson (virtual field trip and component identification).
- Present the "Challenge" question, prompting students to think critically about the geographical and climatic factors that influence the choice of energy sources for their own city.
- Facilitate a discussion on the factors that need to be considered when implementing sustainable energy solutions in different locations.

Expected From Students:

- Review the provided table and solidify their understanding of different sustainable city components and their descriptions.
- Engage in the "Practice" activity by planning and creating a poster focused on clean energy.
- Apply their knowledge of sustainable energy sources and their benefits in the poster design.
- Consider the "Challenge" question and think critically about how geographical location and climate impact energy choices.
- Participate in a discussion, sharing their ideas about suitable energy sources for their city based on environmental factors.

Materials Needed:

- Student books with Page 3 of Lesson 3.
- Poster-making materials (large paper, markers, colored pencils, etc.).
- (Optional) Access to online resources or images related to clean energy.

Step-by-Step Instructions:

1. Review the Table (10-15 minutes):

- Direct students to the table at the top of Page 3.
- Briefly review each component and its description, allowing students to ask any clarifying questions. This reinforces the concepts learned on the previous page.

2. Introduce "Practice" - Poster Creation (20-25 minutes):

- Explain the "Practice" activity: students need to create a poster about clean energy.
- Discuss the purpose of a poster (to inform and persuade).
- Encourage them to include:
 - A clear title related to clean energy.
 - Visual elements (drawings, diagrams, etc.).
 - Information about different types of clean energy (referencing the table or their own research).
 - Benefits of using clean energy for a sustainable city.
- Provide the poster-making materials and allow students time to work individually or in small groups.

3. Present the "Challenge" Question (10 minutes):

- Direct students' attention to the "Challenge" question:
"Which types of energy you can apply in your city (depend on geographical location and climate)?"
- Explain that this question requires them to think specifically about their own local context.
- Briefly discuss the concepts of geographical location (e.g., coastal, mountainous, flat) and climate (e.g., sunny, windy, rainy) and how these might influence the availability and effectiveness of different energy sources.

4. Facilitated Discussion on the "Challenge" (15-20 minutes):

- Lead a class discussion where students share their ideas about suitable energy sources for their city.
- Prompt them with questions like:
 - "Does your city get a lot of sunlight? If so, how could solar energy be used?"
 - "Is your area known for strong winds? How could wind energy be harnessed?"
 - "Are there rivers or other water bodies nearby? Could hydropower be an option?"
 - "What other geographical or climatic factors might make certain energy sources more or less suitable?"
- Encourage them to justify their choices based on their understanding.

Objectives:

- 1-Recognize how wind turbines transform wind energy into electrical energy.
- 2-Understand that wind turbine efficiency is affected by blade design and wind speed.
- 3-Explain the environmental benefits of using wind power as a renewable energy source.
- 4-Identify challenges in connecting wind energy to the power grid.
- 5-Consider different clean energy sources for a sustainable city beyond wind power.

Challenge Questions of the Lesson

- 1-How is energy changed in a wind turbine?
- 2-What makes a wind turbine work better?
- 3-Why is wind energy good for the environment?
- 4-What makes it hard to use wind power all the time?

☐ What other clean energies could our city use?

SEL relation

Self-Awareness: Students reflect on their understanding of energy and its sources.

Critical Thinking: Students compare energy sources and analyze challenges.

Creativity: Students brainstorm different clean energy options.

Reflection: Students assess their own learning against the lesson objectives.

Energy change in turbine: Wind makes it spin, which makes electricity.

Better turbine: Good blade shape and strong wind.

Wind good for environment: It doesn't pollute the air.

Hard to use wind power: Wind doesn't blow all the time.

Other clean energies: Sun, water, heat from the ground.

1. Stepping stone & Let's think

Teacher Role:

- Introduce the focus of this lesson: energy transformation, particularly within the context of wind turbines as a renewable energy source for sustainable cities.
- Briefly review the necessary prior knowledge: the definition of energy, the Law of Conservation of Energy, and the role of mechanical solutions in sustainable cities.
- Clearly state the learning outcomes for this lesson, emphasizing the understanding of energy transformation in wind turbines, factors affecting their efficiency, and the environmental benefits of renewable energy like wind power.
- Use the "Let's Think" questions to stimulate students' thinking about energy choices and planning for their smart city project.

Expected From Students:

- Reflect on their existing knowledge of energy, its conservation, and mechanical solutions in sustainable cities.
- Understand that this lesson will focus on how energy is changed from one form to another in wind turbines.
- Recognize that they will learn about what makes wind turbines efficient and the environmental advantages of using wind energy.
- Begin to consider the energy needs of their smart city and potential clean energy sources, addressing the questions posed in the "Let's Think" section.

Materials Needed:

- Student books with Page 1 of Lesson 4.
- Whiteboard or projector to facilitate discussion.

Step-by-Step Instructions:

1. **Review Prior Knowledge (10 minutes):**
 - Begin by briefly discussing the "You need to know..." section. Ask students to recall:
 - The definition of energy and provide examples.
 - The Law of Conservation of Energy (energy cannot be created or destroyed, only transformed).
 - Examples of mechanical solutions used in sustainable cities (e.g., efficient transportation systems, waste sorting mechanisms).
2. **Introduce Learning Outcomes (5 minutes):**
 - Clearly explain the "You will be able to..." statements, highlighting what students will learn about wind energy in this lesson.
3. **Engage with "Let's Think" (15-20 minutes):**
 - Use the "Let's Think" questions to prompt a class discussion about energy planning for their smart city:

- "What are some essential elements that would require energy in a smart city?" (e.g., lighting, transportation, buildings, industries).
 - "What are some energy sources that are considered cleaner than traditional fossil fuels? What might be some of the costs associated with them (both environmental and economic)?"
 - "What are some initial ideas for how we might generate electricity for our smart city? Should we rely on one source or a mix?"
4. **Introduce the Focus on Wind Energy (5 minutes):**
- o Explain that this lesson will specifically delve into wind turbines as a key example of a renewable energy technology and how they transform energy.

Differentiation:

- **Support:** Provide students with visual aids illustrating energy transformation and different types of energy sources. Offer simplified explanations of the Law of Conservation of Energy.
 - **Extension:** Encourage students to research different types of renewable energy beyond wind power and their potential applications in smart cities.
-

2.Explore

Teacher Role:

- Introduce the "Explore" activities as opportunities for students to engage with the concept of energy transfer through an interactive game and to learn more about wind energy as a key renewable resource.
- Guide students on how to access the game, video, and article using the provided links and QR codes.
- Encourage active participation in the energy transfer game and thoughtful engagement with the information presented in the video and article about wind energy.
- Emphasize that these resources will help them deepen their understanding of the learning outcomes for this lesson.

Expected From Students:

- Access and play the energy transfer game using the provided link or QR code.
- Observe and understand the different forms of energy and how they are transformed in the game.

- Watch the video about energy sources for a city and identify various renewable and non-renewable options.
- Read the provided text or the linked article about wind energy and understand its benefits as a sustainable energy source.
- Make connections between the concepts explored in the game, video, and reading material and the learning objectives of the lesson, particularly regarding energy transformation in wind turbines and the advantages of renewable energy.

Materials Needed:

- Student books with Page 2 of Lesson 4.
- Devices with internet access and QR code scanners.
- Projector or screen to display the game, video links, and reading material if used as a class activity.

Step-by-Step Instructions:

- 1. Introduce the "Explore - Do it" Activity (10 minutes):**
 - o Direct students to the "Explore" section and explain that they will be starting with an interactive game to visualize energy transfer.
 - o Provide instructions on how to access the game using the website link or QR code.
 - o Encourage them to pay attention to the different forms of energy and how they change during the game.
- 2. Facilitate Game Play (Time Varies):**
 - o Allow students sufficient time to play the energy transfer game individually or in pairs.
 - o Circulate and offer assistance as needed, prompting students to discuss what they are observing about energy transformation.
- 3. Introduce "Watch it.." Resource (5 minutes):**
 - o Point out the video link and QR code about energy sources for a city.
 - o Suggest that students watch this video to broaden their understanding of different energy options, including

renewables. This can be done as a homework assignment or in class if time allows.

4. Introduce "Read about it" Resource (5 minutes):

- o Direct students to the text about wind energy and the link/QR code to the article.
- o Explain that this resource provides more specific information about wind energy as a sustainable and environmentally friendly option. Encourage them to read it to prepare for further discussions on wind turbines. This can also be assigned as homework or done in class.

5. Connect to Learning Objectives (10 minutes - after exploration):

- o After students have had time to engage with the resources, lead a class discussion to connect their experiences to the learning objectives:
 - "How did the game illustrate the concept of energy transformation?"
 - "What are some of the benefits of wind energy highlighted in the reading material?"
 - "How might wind energy fit into our plans for a sustainable smart city?"

Connecting to STEAM:

- **Science:** Directly exploring the concept of energy transformation through an interactive medium.
- **Technology:** Utilizing digital resources like games and videos for learning.
- **Environmental Science:** Focusing on the benefits of renewable energy sources like wind power.

Differentiation:

- **Support:** Provide students with guiding questions to focus their attention while playing the game, watching the video, and reading the article. Offer summaries of the key concepts if needed.
 - **Extension:** Encourage students to research other interactive simulations or articles about different types of renewable energy and share their findings with the class.
-

3.Focus & Practice :

Teacher Role:

- Use the "Focus" questions to encourage deeper thinking and analysis about wind energy in comparison to other renewable sources and the practical challenges of integrating it into existing power systems.
- Facilitate a discussion based on these questions, drawing upon information from the "Explore" section and students' prior knowledge.
- Introduce the "Practice" activity: building a DIY wind turbine. Emphasize the hands-on application of the concepts learned about wind energy and energy transformation.
- Provide guidance and resources for the DIY wind turbine construction.

Expected From Students:

- Engage in critical thinking to compare the efficiency of wind turbines with other renewable energy sources.
- Understand the challenges associated with integrating wind energy into the electrical grid (e.g., intermittency, storage) and potential solutions.
- Plan and construct a basic DIY wind turbine model capable of generating a small amount of electricity (if feasible with available materials).

- Apply their understanding of energy transformation and wind power principles in the design and construction of their DIY turbine.

Strategies to use with this part:

1. Comparative Analysis (Focus Questions)

- **Strategy:** Encourage students to compare and contrast wind energy with other renewable sources.
- **Implementation:** Facilitate a class discussion using a Venn diagram or a comparison chart on the board. Ask students to brainstorm the advantages and disadvantages of wind, solar, and hydroelectric power in terms of efficiency, reliability, environmental impact, and cost.

2. Problem-Solving Scenario (Grid Integration)

- **Strategy:** Present the challenge of integrating wind energy into the grid as a problem to be solved.
- **Implementation:** Divide students into small groups and assign each group a specific challenge related to wind energy integration (e.g., intermittency, storage, transmission). Have them brainstorm potential solutions and share their ideas with the class.

3. Hands-on Inquiry (DIY Wind Turbine)

- **Strategy:** Use the DIY wind turbine project as an opportunity for experiential learning and inquiry-based science.
- **Implementation:** Encourage students to experiment with different blade designs, materials, and generator setups. Guide them to test their turbines and observe how changes affect the amount of electricity produced. Ask guiding questions like, "What happens if you change the angle of the blades?" or "How does the size of the blades affect the output?"

4. Design Thinking (DIY Wind Turbine)

- **Strategy:** Frame the DIY project using the principles of design thinking.

- o **Implementation:** Guide students through the stages of understanding the goal (generating electricity from wind), ideating different designs, prototyping their turbines, testing their prototypes, and iterating on their designs for improvement.
5. **Visual Learning (DIY Inspiration)**
- o **Strategy:** Provide visual resources and examples of different DIY wind turbine designs.
 - o **Implementation:** Show images or short videos of various simple wind turbine models to inspire students' creativity and provide ideas for construction techniques and materials.
6. **Collaborative Building (DIY Wind Turbine)**
- o **Strategy:** Encourage teamwork and collaboration during the construction of the DIY wind turbines.
 - o **Implementation:** Have students work in small groups, assigning different roles and responsibilities to each member. Encourage them to share ideas, troubleshoot problems together, and learn from each other's approaches.

Materials Needed:

- Student books with Page 3 of Lesson 4.
- Whiteboard or projector for discussion.
- Materials for the DIY wind turbine project (e.g., small DC motors/generators, cardboard, plastic bottles, wire, LEDs, multimeters).
- (Optional) Online resources or instructions for DIY wind turbine projects.

Step-by-Step Instructions:

1. **Engage with "Focus" Questions (20-25 minutes):**
 - o Begin by addressing the first "Focus" question: "How does the efficiency of wind turbines compare to other renewable energy sources, such as solar or hydroelectric power?"

- Facilitate a discussion, prompting students to recall information about the efficiency of different renewable sources from their prior knowledge or the "Explore" resources. Guide them to understand that efficiency can vary greatly depending on factors like technology, location, and conditions.
 - o Move on to the second question: "What challenges arise when integrating wind energy into the electrical grid, and how can they be addressed?"
 - Lead a discussion about the intermittent nature of wind power, the need for energy storage solutions (batteries, pumped hydro), and the development of smart grids to manage fluctuating power supply.
2. **Introduce "Practice" - DIY Wind Turbine (Time Varies - can be a multi-day activity):**
- o Introduce the hands-on project: building a DIY wind turbine to generate electricity.
 - o Explain the goal is to create a model that can capture wind energy and convert it into electrical energy, even if it's just enough to light a small LED or produce a measurable voltage.
 - o Provide students with basic instructions or guide them to find resources for DIY wind turbine projects.
 - o Discuss the materials they will need and safety precautions.
 - o Allow sufficient time for planning, construction, and testing of their wind turbines. This activity can be done individually or in small groups.

Differentiation:

- **Support:** Provide students with simplified instructions, pre-cut materials, or templates for their DIY wind turbines. Offer more guidance during the construction process.
- **Extension:** Challenge students to optimize their wind turbine designs for maximum energy output, measure the voltage and current produced, or research different types of wind turbine designs.

4.Challenge:

“Leave it to students search and creativity “

Teacher Role:

- Present the "Challenge" to expand students' thinking beyond wind energy and consider other clean energy options for their sustainable city.
- Guide students to connect their learning back to the Engineering Design Process (EDP), specifically the "Create" stage, as prompted by the robot.
- Encourage students to reflect on their understanding of the lesson's learning outcomes through the "Now I can..." self-assessment.

Expected From Students:

- Think creatively about different clean energy sources besides wind power that could be integrated into their sustainable city design.
- Understand the cyclical nature of the EDP and the importance of revisiting earlier stages as their understanding evolves.
- Reflect on their learning and assess their ability to recognize energy transformation in wind turbines, understand their efficiency, and grasp the environmental benefits of renewable energy like wind power.
- Check the "Now I can..." statements that they feel confident about.

Materials Needed:

- Student books with Page 4 of Lesson 4.
- (Optional) Whiteboard or projector for brief discussion.
- Student Engineering Design Process (EDP) documentation.

Step-by-Step Instructions:

1. **Present the "Challenge" (5-10 minutes):**
 - o Direct students to the "Challenge" question and encourage them to brainstorm other clean energy sources suitable for a

city (e.g., solar, geothermal, hydro). Briefly discuss how they might integrate these into their sustainable city plans.

2. Connect Back to the EDP (5 minutes):

- o Refer to the robot's prompt and the EDP diagram. Emphasize that as they learn more (like about energy sources), they should revisit their EDP, particularly the "Create" stage, to incorporate these new ideas into their designs.

3. "Now I can..." Self-Assessment (5 minutes):

- o Have students read the "Now I can..." statements and individually check the boxes for the learning outcomes they feel they have achieved during this lesson.

Differentiation:

- **Support:** Provide students with a list of clean energy sources to consider for the "Challenge." Briefly review the EDP stages if needed.
- **Extension:** Encourage students to research the feasibility and integration challenges of different clean energy sources in urban environments.

Objectives:

- 1-Imagine and design their own sustainable city using Tinkercad as a visual planning tool.
- 2-Identify smart features that make cities environmentally friendly, efficient, and comfortable for people.
- 3-Apply critical thinking and creativity to improve city designs using renewable energy, green spaces, and smart waste and water systems.
- 4-Reflect on their design choices and communicate how these features solve real-world sustainability challenges.

Challenge Questions of the Lesson

- 1-What does a smart and sustainable city need in order to help people and the planet at the same time?
- 2-How can you use design tools like Tinkercad to plan and share your ideas for a better city?
- 3-Why is it important to include features like green spaces, renewable energy, and smart water systems in a future city?
- 4-How do the choices we make today when designing cities affect the way people live tomorrow?
- 5-What other clean energies could our city use?

SEL relation

Self-awareness: Students reflect on their own ideas for solving real-world problems by creating thoughtful city designs.

Responsible decision-making: Students explore the impact of their design choices on the environment and community well-being.

Collaboration and communication: Students discuss, share, and exchange ideas about sustainable living and learn to value the opinions of others.

Empathy and global citizenship: Students develop an understanding of how sustainable cities help all people live healthier, safer, and more comfortable lives.

Here are short and simple answers :

- 1-It needs clean energy, green spaces, smart buildings, and systems that save water and reduce waste.
- 2-You can build 3D models to show your ideas clearly and make changes easily.
- 3-These features help protect nature, save resources, and make life better for everyone.
- 4-Smart choices today create safer, cleaner, and more comfortable places to live in the future.

1. Engage:

Teacher Role:

- Introduce the topic of sustainable cities by connecting it to students' real-life experiences with city living, parks, neighborhoods, and energy use.
 - Use interactive strategies such as thought-provoking visuals, real-world examples, and open-ended questions to spark curiosity and activate prior knowledge.
 - Encourage students to make predictions about city design challenges, especially focusing on energy systems, green spaces, and sustainability.
 - Facilitate a supportive discussion space where students feel confident to share their ideas, ask questions, and explore different design solutions.
 - Use scaffolding techniques like visual aids, sentence starters, and guiding questions to help all students actively participate and share their perspectives.
-

Expected From Students:

- Participate actively by discussing their ideas about cities, energy, and green spaces.
- Engage with visuals and prompts to build curiosity about what makes a city sustainable.
- Share personal experiences of visiting parks, seeing solar panels, or noticing city design features like bike lanes or recycling systems.
- Think critically about the benefits and challenges of integrating renewable energy and nature-friendly spaces in cities.

- Make predictions about the challenges involved in planning an eco-friendly city.
-

Strategies to use for implementation:

Preview with a Hook:

Strategy: Begin with an intriguing question or real-world comparison.

Implementation:

- Show two contrasting images: one of a dense city skyline with lots of traffic and smog, and another of a city full of parks, bike lanes, and solar panels.
 - Ask students:
“Which of these cities do you think is healthier to live in? Why?”
 - Guide a brief discussion connecting their ideas to the day's topic: **designing a sustainable city.**
-

Make Predictions:

Strategy: Use the lesson title to spark curiosity.

Implementation:

- Write the lesson title on the board: **“Design Your City”**
 - Ask:
“If you were in charge of building a new city, what would you include to make sure it’s good for both people and nature?”
 - Jot down predictions on the board and refer back to them during or after the lesson.
-



Set the Scene with a Visual:

Strategy: Use cityscape images to introduce the topic.

Implementation:

- Show images of real-world cities that use renewable energy and green spaces (examples: Copenhagen, Singapore, or Masdar City in the UAE).
 - Ask:
“What features do you notice in these cities? How do they help the environment and the people who live there?”
-



City Design Walk (Adapted from Story Walk):

Strategy: Create an interactive gallery of sustainable city elements.

Implementation:

- Set up stations around the room with pictures and short descriptions of city features:
(solar panels, wind turbines, bike paths, recycling bins, rooftop gardens).
 - Ask students at each station:
“How does this help a city become more sustainable?”
 - Encourage pairs or small groups to rotate and discuss their ideas.
-



Think-Pair-Share:

Strategy: Foster collaboration and idea sharing.

Implementation:

- Ask:
“What are some problems cities face when trying to use solar or wind energy?”
- Have students think individually, then discuss with a partner.

- Invite pairs to share their answers with the class.
-

Interactive Exploration:

Strategy: Keep students engaged with hands-on questioning.

Implementation:

- Use guiding questions as students explore city design models or drawings.
 - Example:
“Where would you place wind turbines in your city to get the most energy?”
“How will you make space for green parks while still fitting buildings and roads?”
 - Encourage students to reflect on their choices and revise their city designs.
-

2. Explore:

Teacher Role:

- Encourage students to visualize their own version of a sustainable city by prompting them to think creatively about what features and systems it might include.
- Guide students in using **Tinkercad** to turn their ideas into a simple 3D model, which they will later use as a reference for more detailed design and problem-solving.
- Support students as they explore the **Smart City** video by helping them connect the concepts in the video to the elements they want to include in their own city designs.
- Ask open-ended questions such as:
“Why did you choose to add solar panels or wind turbines?”
“How do parks and green areas help a city?”

to deepen their reasoning and reflections during the modeling process.

- Encourage collaboration by inviting students to share their ideas and compare different approaches to sustainable city design.
-

Expected From Students:

- Use creative thinking to imagine what a sustainable city could look like and what features it needs to support both people and nature.
 - Build a simple 3D city design using **Tinkercad**, including elements such as renewable energy systems, green spaces, transportation systems, and water-saving solutions.
 - Watch the linked **Smart City** video and take note of new ideas or technologies that they might want to add to their own designs.
 - Communicate their choices by explaining the reasons behind the features they chose for their city.
 - Reflect on the design process and be open to improving their models based on new insights or feedback from peers and the teacher.
-

Strategies to use for implementation:



Visual Brainstorming:

Strategy: Use drawings and quick sketches before moving to Tinkercad.

Implementation:

- Have students sketch a rough map of their sustainable city on paper before modeling it.

- Encourage them to label features such as solar panels, green roofs, bike paths, water recycling systems, and public transport.
-

Guided Questioning:

Strategy: Use questions to push deeper thinking during design.

Implementation:

- Ask questions like:
“Where will people get clean energy in your city?”
“How will you reduce waste and pollution?”
 - Let students explain their reasoning as they work on their Tinkercad model.
-

Video Reflection:

Strategy: Use the suggested video as a knowledge spark.

Implementation:

- Play the **Smart City** video linked in the lesson.
 - After watching, ask:
“Which idea from the video would you include in your city, and why?”
 - Allow students to add new elements inspired by the video to their Tinkercad designs.
-

Peer Feedback:

Strategy: Promote collaboration and constructive feedback.

Implementation:

- Once students complete their first version of the city design, pair them up.

- Let each pair review each other's city in Tinkercad, asking:
“What part of this city helps the environment the most?”
“Is there anything you’d suggest adding or changing?”
-

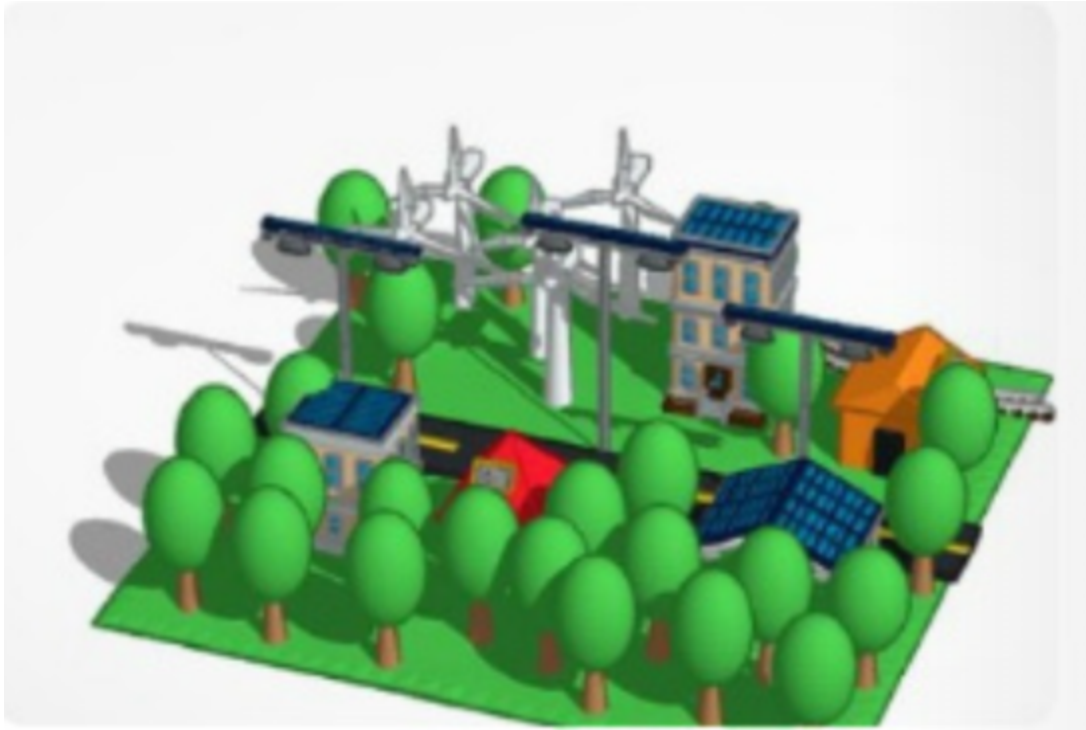
3. Assessment:

Teacher Role:

- Guide students in reviewing their **Tinkercad city model** and encourage them to reflect on the smart and sustainable features they included.

LET YOUR STUDENTS BE CREATIVE

- Ask students to highlight the specific **changes or improvements** they plan to make to turn their city into a smart and eco-friendly place.
- Help them revisit their **EDP (Engineering Design Process) worksheet** to update the **Plan** section based on what they learned from designing and choosing building materials.
- Support self-reflection by prompting students to explain how their changes address real-world sustainability challenges like clean energy, green spaces, or efficient water use.
- Encourage students to complete the **“Now I can...”** checklist to help them reflect on their learning and feel proud of their progress.



Expected From Students:

- Identify and highlight smart solutions and improvements that will make their city more sustainable and efficient.
 - Apply the knowledge gained from the lesson by updating their EDP worksheet, especially the **Plan** section, to show how they will apply their chosen materials and smart ideas.
 - Complete the “**Now I can...**” checklist, demonstrating their understanding of key concepts like green spaces, water and waste systems, and integrated renewable energy.
 - Communicate their choices clearly, using reasoning to explain why they believe their solutions are both smart and sustainable.
 - Show growth in critical thinking and creativity through both their design choices and written reflections.
-

Strategies to use for implementation:

Feature Highlighting:

Strategy: Focus students on the specific parts of their city design that make it “smart.”

Implementation:

- Ask students to mark or name at least 3 changes they’ve made that address sustainability.
 - Example questions:
"Which feature of your city helps save energy?"
"How does your design handle waste and water wisely?"
-

EDP Reflection:

Strategy: Update the EDP worksheet with new ideas.

Implementation:

- After selecting the best building materials and smart features, have students write or sketch updates in the **Plan** section of their EDP.
 - Discuss how their choices fit real-life engineering challenges.
-

Now I Can Checklist:

Strategy: Use the checklist as a self-assessment and discussion starter.

Implementation:

- Let students read and tick off each item as they confirm their understanding.
- Invite volunteers to share examples of their design choices for each point, encouraging peer learning.

Objectives:

- 1-Understand smart lighting systems' role in energy efficiency.
- 2-Build and test prototypes using motion or sound sensors.
- 3-Collaborate effectively and reflect on real-world applications.

Challenge Questions of the Lesson

- 1-Why are smart lighting systems better than traditional ones?
- 2-How do sensors save energy, and how could this technology be scaled for public spaces?
- 3-What improvements can you make to your prototypes?

SEL relation

Self-awareness: Students reflect on their designs for smart lighting systems and consider how their idea

Responsible decision-making: Students evaluate the environmental and societal impact of their smart lighting choices.

Collaboration and communication: Students share and discuss innovative ideas in groups, appreciating diverse perspectives.

Empathy and global citizenship: Students understand how energy-efficient solutions improve the quality of life for communities worldwide.

1.Engage

Teacher Role:

- Introduce the concept of smart cities and their role in energy efficiency through an engaging discussion.
- Share real-world examples, such as energy-saving technologies in modern buildings, to connect with students' prior knowledge.
- Use interactive strategies like showing visuals of sensors, LED lighting, and load balancing in action.
- Pose open-ended questions like, *"What would make your ideal city smart?"* to spark curiosity and discussion.
- Guide students to consider the benefits of automation and its impact on daily life.

Expected From Students:

- Actively participate by sharing thoughts and asking questions about smart city features.
- Analyze visuals and examples to draw meaningful connections.
- Predict outcomes of using energy-saving solutions and discuss ideas with peers.
- Reflect on the advantages of automation for personal and societal benefits.

Strategies to use with this part

1.Preview with a Hook:

- Strategy: Start with a provocative question or striking visual.
- Implementation:
 - Display images of a smart building equipped with sensors vs. a traditional building.

- Ask: *"Which building is more energy-efficient and why?"*
- Discuss briefly to link concepts with the lesson objectives.

2. Make Predictions:

- o Strategy: Introduce the question from the "Let's Think" section.
- o Implementation:
 - Show the question: *"How can the use of LED lighting, occupancy sensors, and load balancing reduce energy consumption?"*
 - Have students write or share predictions, which can be revisited later during the lesson.

3. Interactive Storytelling:

- o Strategy: Illustrate the concept of automation creatively.
 - o Implementation:
 - Use props to mimic the operation of sensors in a smart city.
 - Ask students to imagine and narrate scenarios where automation saves energy (e.g., motion sensors turning off unused lights).
-

2.Explore (DIY Project)

Teacher Role:

- Introduce the DIY project by explaining its purpose: designing and creating a prototype of a smart lighting system.
- Encourage students to think critically about energy efficiency and automation while emphasizing the real-world impact of their designs.
- Provide clear instructions and a list of materials required for the project.
- Facilitate the process by guiding students through brainstorming, planning, and creating their prototypes step by step.

- Offer support and feedback throughout the activity to ensure students stay engaged and inspired.
- Guide students through the "Watch it" and "Read about it" sections to gather ideas and deepen their understanding of the topic.

Expected From Students:

- Actively participate by brainstorming ideas for energy-efficient smart lighting systems.
- Collaborate with peers to plan and execute their DIY projects.
- Use creativity and problem-solving skills to design and build their prototypes.
- Reflect on how their designs could be implemented in real-world scenarios to promote energy savings.

Strategies to use with this part

1. Materials Preparation:

- Strategy: Provide a list of materials for the DIY activity.
- Implementation:
 - Include items like cardboard, LEDs, batteries, wires, sensors, and switches.
 - Suggest that students bring additional materials to personalize their prototypes.

2. Step-by-Step Guide:

- Strategy: Break down the DIY project into manageable steps.
- Implementation:
 - Step 1: Brainstorm ideas for a smart lighting system.
 - Step 2: Sketch the design on paper, highlighting key features like sensors and automated controls.
 - Step 3: Build the prototype using provided materials.
 - Step 4: Test the prototype to ensure functionality and energy efficiency.

3. Collaborative Creation:

- o Strategy: Encourage teamwork and shared creativity.
- o Implementation:
 - Divide students into small groups to work on their prototypes.
 - Provide prompts like *"How can your system detect occupancy to save energy?"* to inspire ideas.

4. Presentation and Reflection:

- o Strategy: Allow students to showcase their prototypes.
 - o Implementation:
 - Each group presents their design and explains how it saves energy.
 - Facilitate a discussion on challenges faced during the project and lessons learned.
-

5. Watch It:

- o Strategy: Encourage students to explore the provided video link.
- o Implementation:
 - Share the QR code and link.
 - Have students observe real-world applications of smart lighting.
 - Ask reflective questions like *"How do sensors and automation in the video contribute to energy efficiency?"*

6. Read About It:

- o Strategy: Facilitate understanding through the article.
 - o Implementation:
 - Provide the QR code and link.
 - Highlight key points about energy-saving features and smart lighting.
 - Encourage students to jot down ideas inspired by the article for their designs.
-

3.Evaluate

Teacher Role:

- Facilitate the assessment by guiding students through the "Focus" section.
- Discuss the reasoning behind each correct answer to deepen understanding of smart lighting technologies.
- Reinforce key concepts while addressing misconceptions.

Expected From Students:

- Thoughtfully answer the multiple-choice questions, showing comprehension of the lesson's topics.
- Reflect on their answers and participate in discussions to explore related concepts.

Strategies to use with this part

1. Clarify

Concepts:

- Strategy: Discuss the terms and technologies mentioned in the questions.
- Implementation:
 - **Question 1:** What is the primary technology used in smart lighting bulbs?
 - Correct Answer: *LED*.
 - Explanation: LED technology is the foundation for smart lighting bulbs due to their energy efficiency, long life, and adaptability to smart features.
 - **Question 2:** Which feature is not typically part of a smart lighting system?
 - Correct Answer: *Sound amplification*.
 - Explanation: Smart lighting systems focus on automation and energy efficiency, not audio capabilities.
 - **Question 3:** How can smart lights be connected for automated operation?
 - Correct Answer: *d. Both B & C (Wi-Fi and Bluetooth)*.

- Explanation: Smart lights often use Wi-Fi and Bluetooth for connectivity, enabling integration with smart home systems and remote control.

2. Engage in Discussion:

- o Strategy: Facilitate group discussions on each question.
- o Implementation:
 - Ask reflective questions like *"Why is LED lighting the preferred technology for energy efficiency?"* and *"How does Wi-Fi and Bluetooth enhance the functionality of smart lights?"*

3. Extend the Learning:

- o Strategy: Connect the assessment to real-world applications.
 - o Implementation:
 - Challenge students to think about the scalability of smart lighting systems in city infrastructure.
 - Pose questions like *"How can we apply smart lighting technologies to public spaces like parks or streets?"*
-

Practice & Challenge:

Teacher Role:

- Guide students through the **Practice** activity by explaining how to design a functional smart lighting system with sound or motion sensors.
- Encourage creativity and problem-solving as students take on the **Challenge** to create a laser alarm system.
- Reinforce the learning outcomes listed in the **Now I can...** section to highlight what students have achieved and how they can apply their knowledge.
- Provide feedback and support as students build and test their systems.

Expected From Students:

- Apply their knowledge of sensors, circuits, and automation to build a smart lighting system.
- Demonstrate engineering and critical thinking skills in creating a prototype laser alarm system for home security.

- Reflect on their achievements and understanding of the key concepts covered in the lesson.

Strategies to use with this part

1. Practice:

- o Strategy: Walk students through creating a smart lighting system with sound or motion sensors.
- o Implementation:
 - Provide materials such as LEDs, sound or motion sensors, breadboards, and connecting wires.
 - Guide them to assemble the circuit, ensuring the sensors activate the lights automatically.
 - Encourage troubleshooting to refine their systems for reliable operation.

2. Challenge:

- o Strategy: Introduce the laser alarm system as an optional but engaging advanced project.
- o Implementation:
 - Provide materials such as laser pointers, mirrors, photoresistors, and buzzers.
 - Help students create a basic circuit where the laser beam, when interrupted, triggers the buzzer.
 - Discuss real-world applications, such as using this system in security setups.

3. Reflection - "Now I can...":

- o Strategy: Help students connect their learning to the listed outcomes.
- o Implementation:
 - Facilitate a discussion where students share which skills or concepts they improved the most.

- Ask questions like, *"Which part of building the smart light system or laser alarm was most challenging, and how did you solve it?"*
 - Highlight how these projects demonstrate their understanding of sensors, circuits, and automation.
-

Activities details for teachers

Control LED By Clap Using Arduino and Sound Sensor

Things used in this project

Hardware components



Arduino UNO

× 1

SparkFun KY-038 Sound Sensor

× 1



LED (generic)

× 1



Jumper wires (generic)

Male To Female

× 5

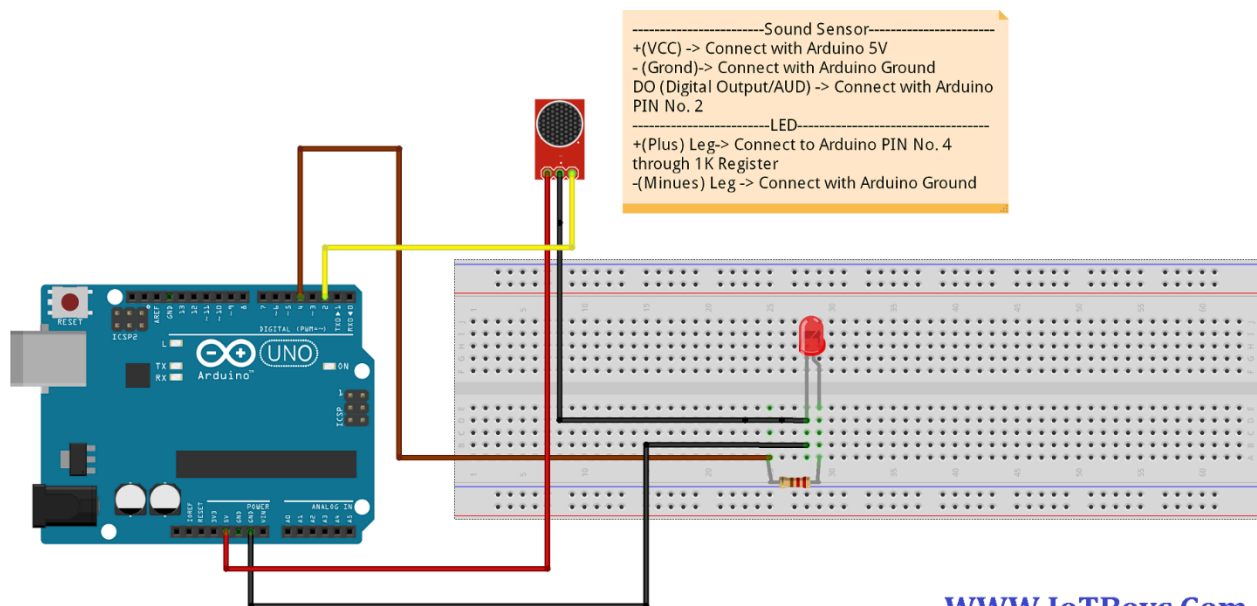


Resistor 1k ohm

× 1



Breadboard (generic)



WWW.IoTBoys.Com

Code:

```
int soundSensor=2;  
int LED=4;  
boolean LEDStatus=false;
```



```

void setup() {
  pinMode(soundSensor,INPUT);
  pinMode(LED,OUTPUT);
}

void loop() {

  int SensorData=digitalRead(soundSensor);
  if(SensorData==1){

    if(LEDStatus==false){
      LEDStatus=true;
      digitalWrite(LED,HIGH);
    }
    else{
      LEDStatus=false;
      digitalWrite(LED,LOW);
    }
  }
}
}

```

For more details , check this link:

<https://www.hackster.io/iotboys/control-led-by-clap-using-arduino-and-sound-sensor-e31809>

Motion Sensor with LED – Arduino Tutorial

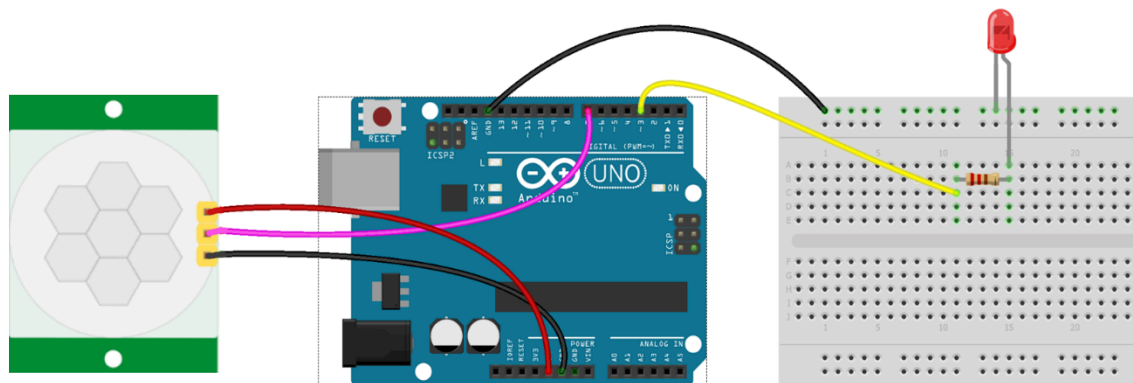
Hardware Components

You will require the following hardware for Motion Sensor with Arduino.

S.no	Component	Value	Qty
1.	<u>Arduino UNO</u>	–	1
2.	USB Cable Type A to B	–	1
3.	Motion Sensor	<u>HC-SR50</u> <u>1</u>	1

S.no	Component	Value	Qty
4.	Power Adapter for Arduino	9V	1
5.	<u>LED</u>	—	1
6.	<u>Resistor</u>	Ω	1
7.	Breadboard	—	1
8.	Jumper Wires	—	1

Motion Sensor Arduino Circuit



For Complete Details Visit :
www.Circuits-DIY.com

Code:

```
const int MOTION_SENSOR_PIN = 7; // Arduino pin connected to the
OUTPUT pin of motion sensor
```

```
const int LED_PIN = 3; // Arduino pin connected to LED's pin
```

```
int motionStateCurrent = LOW; // current state of motion sensor's
pin
```

```
int motionStatePrevious = LOW; // previous state of motion sensor's
pin
```

```
void setup() {  
    Serial.begin(9600);           // initialize serial  
    pinMode(MOTION_SENSOR_PIN, INPUT); // set arduino pin to  
input mode  
    pinMode(LED_PIN, OUTPUT);     // set arduino pin to output mode  
}  
  
void loop() {  
    motionStatePrevious = motionStateCurrent; // store old state  
    motionStateCurrent = digitalRead(MOTION_SENSOR_PIN); // read  
new state  
    if (motionStatePrevious == LOW && motionStateCurrent == HIGH) {  
// pin state change: LOW -> HIGH  
        Serial.println("Motion detected!");  
        digitalWrite(LED_PIN, HIGH); // turn on  
    }  
    else  
        if (motionStatePrevious == HIGH && motionStateCurrent == LOW) {  
// pin state change: HIGH -> LOW  
            Serial.println("Motion stopped!");  
            digitalWrite(LED_PIN, LOW); // turn off  
        }  
    }  
}
```

```

1.  const int MOTION_SENSOR_PIN = 7;    // Arduino pin connected to the motion sensor
2.  const int LED_PIN           = 3;    // Arduino pin connected to LED's pin
3.  int motionStateCurrent      = LOW;   // current state of motion sensor's pin
4.  int motionStatePrevious     = LOW;   // previous state of motion sensor's pin
5.
6.  void setup() {
7.      Serial.begin(9600);              // initialize serial
8.      pinMode(MOTION_SENSOR_PIN, INPUT); // set arduino pin to input mode
9.      pinMode(LED_PIN, OUTPUT);         // set arduino pin to output mode
10. }
11.
12. void loop() {
13.     motionStatePrevious = motionStateCurrent;    // store old state
14.     motionStateCurrent  = digitalRead(MOTION_SENSOR_PIN); // read new state
15.
16.     if (motionStatePrevious == LOW && motionStateCurrent == HIGH) { // pin state
change: LOW -> HIGH
17.         Serial.println("Motion detected!");
18.         digitalWrite(LED_PIN, HIGH); // turn on
19.     }
20.     else
21.         if (motionStatePrevious == HIGH && motionStateCurrent == LOW) { // pin state
change: HIGH -> LOW
22.             Serial.println("Motion stopped!");
23.             digitalWrite(LED_PIN, LOW); // turn off
24.         }
25.     }

```

For more details , check this link:

<https://www.circuits-diy.com/motion-sensor-with-led-arduino-tutorial/>

Laser alarm system

Arduino code

```
int buzzer = 9;
int LDR = A0;
int laserPin = 8;
int sensorValue;

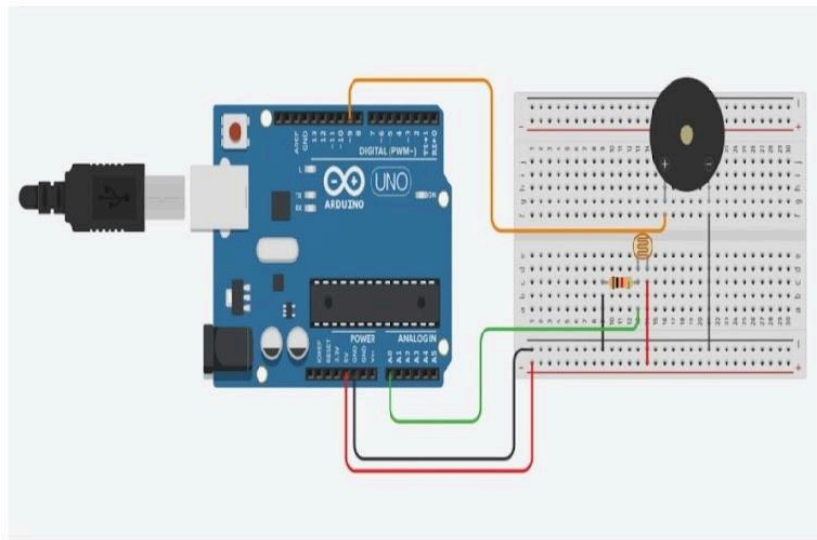
void setup() {
  pinMode(buzzer, OUTPUT);
  pinMode(laserPin, OUTPUT);
}

void loop() {

  digitalWrite(laserPin, HIGH);

  sensorValue = analogRead(LDR);

  // Check if the laser beam is hindered
  if (sensorValue <= 800) {
    // If the beam is hindered, turn on the alarm
    digitalWrite(buzzer, HIGH);
  } else {
    // If the beam is not hindered, turn off the alarm
    digitalWrite(buzzer, LOW);
  }
}
```



Objectives:

- 1-Understand how RFID reads tags.
- 2-Know RFID system parts (tag, reader).
- 3-Build a circuit with RFID and a motor.
- 4-Program Arduino to move a motor with RFID.
- 5-Explain how RFID makes secure doors.

- ☐ Discuss RFID uses in smart cities.

Challenge Questions of the Lesson

- 1-How can we use RFID to build a simple smart door for a more secure city?
- 2-How does the reader get info from the tag without touching?
- 3-Where else is RFID used in smart cities?

SEL relation

Self-Awareness: Think about your own experiences with security and how you felt during the project.

Self-Management: Stay focused while building and coding; don't give up when it's hard.

Social Awareness: Consider how security helps everyone in a city.

Relationship Skills: Share ideas and help others if working together.

Responsible Decision-Making: Think about why security tech is important.

Here are short and simple answers:

- **Smart Door:** Program reader to open a lock only for specific tags.
- **Reader & Tag:** Reader sends radio waves; tag uses them to send back info.
- **RFID in Cities:** Transport pay, library, parking, pet ID, inventory.

1. Engage:

Teacher Role:

- Introduce the concept of automation and security in accessing spaces through an engaging discussion that connects to students' experiences with keys, access cards, or even automated doors they might have encountered (e.g., in malls, buildings).
- Incorporate interactive strategies such as thought-provoking visuals of different access control systems (traditional keys, key cards, fingerprint scanners, automated garage doors), storytelling elements about the convenience and security of automated systems, and open-ended questions about the benefits and drawbacks of each.
- Encourage students to make predictions by prompting them to analyze the "Let's Think" question and the "You will be able to..." objectives, helping them anticipate the role of technology like RFID in creating a smart door.
- Facilitate meaningful discussions by creating a supportive environment where students feel confident to share their experiences with different access methods, ask questions about how automation and security work, and consider the potential of robotics in everyday life.
- Provide targeted support by guiding discussions with scaffolding techniques, such as providing analogies to familiar systems or breaking down the "Let's Think" question into smaller parts, ensuring that all students can participate and contribute their initial ideas about smart access systems.

Expected From Students:

- Actively contribute to discussions by sharing their experiences with different ways of accessing spaces, asking questions about how security and automation are achieved, and making thoughtful predictions about the technologies involved in a smart door.
- Engage with multimedia elements such as visuals of various access control systems and the "Let's Think" prompt to deepen their understanding of the problem and potential solutions.
- Express personal ideas confidently while listening respectfully to peers' experiences and opinions on security and automation, fostering a collaborative and inclusive learning environment.
- Make meaningful connections by reflecting on how technology can simplify and secure everyday tasks, relating the concept of a smart door to broader applications of robotics and automation and beyond.

- Demonstrate critical thinking by analyzing the "Let's Think" question, considering different ways technology could be used for access control, and forming initial hypotheses about the components and programming involved in a smart garage door system.

Strategies to use with this part

Preview with a Hook:

- **Strategy:** Begin with an intriguing question or striking image.
- **Implementation:**
 - Display an image of someone struggling to find their keys in the rain versus an image of a garage door smoothly opening automatically as a car approaches.
 - Ask, “Which of these scenarios seems more convenient and secure? Why?”
 - Discuss briefly to connect students’ ideas with the benefits of automated access systems.
- **Make Predictions:**
 - **Strategy:** Share the lesson title: "Smart Door."
 - **Implementation:**
 - Show the lesson title: "Smart Door."
 - Ask students, “Based on the title and the ‘Let’s Think’ question, what do you predict are some of the technologies that might be used to create a ‘smart’ door, especially for a garage?”
 - Jot down student predictions, perhaps focusing on keywords like "automatic," "secure," and any technologies they might already know.
- **Set the Scene with a Visual:**
 - **Strategy:** Use a picture or short video clip contrasting a traditional keyed lock with a modern smart lock or an automated garage door system.
 - **Implementation:**
 - Show a picture comparing a traditional key and lock to an RFID tag and reader on a garage door.
 - Optionally, play a short clip of an automated garage door opening using a remote or a card.

- Ask students, “What are the key differences you observe between these two ways of accessing a space? What makes the second one ‘smart’?”
- **Think-Pair-Share:**
 - **Strategy:** Facilitate collaborative discussions around the "Let's Think" question.
 - **Implementation:**
 - Ask students to individually think about different technologies that could be used for secure and automated garage access.
 - Have them pair with a partner to share their ideas and discuss the pros and cons of each.
 - Invite volunteers to share their group's ideas with the whole class.
- **Interactive Storytelling (Relating to a Familiar Context):**
 - **Strategy:** Engage students by relating the concept to a familiar place that might use some form of access control.
 - **Implementation:**
 - Ask, “Think about places that have some kind of special way to open doors or gates, like a building entrance or even some parking areas. What kind of systems do they use, and why do you think they use those instead of just regular keys?”

2. Explore:

Teacher Role:

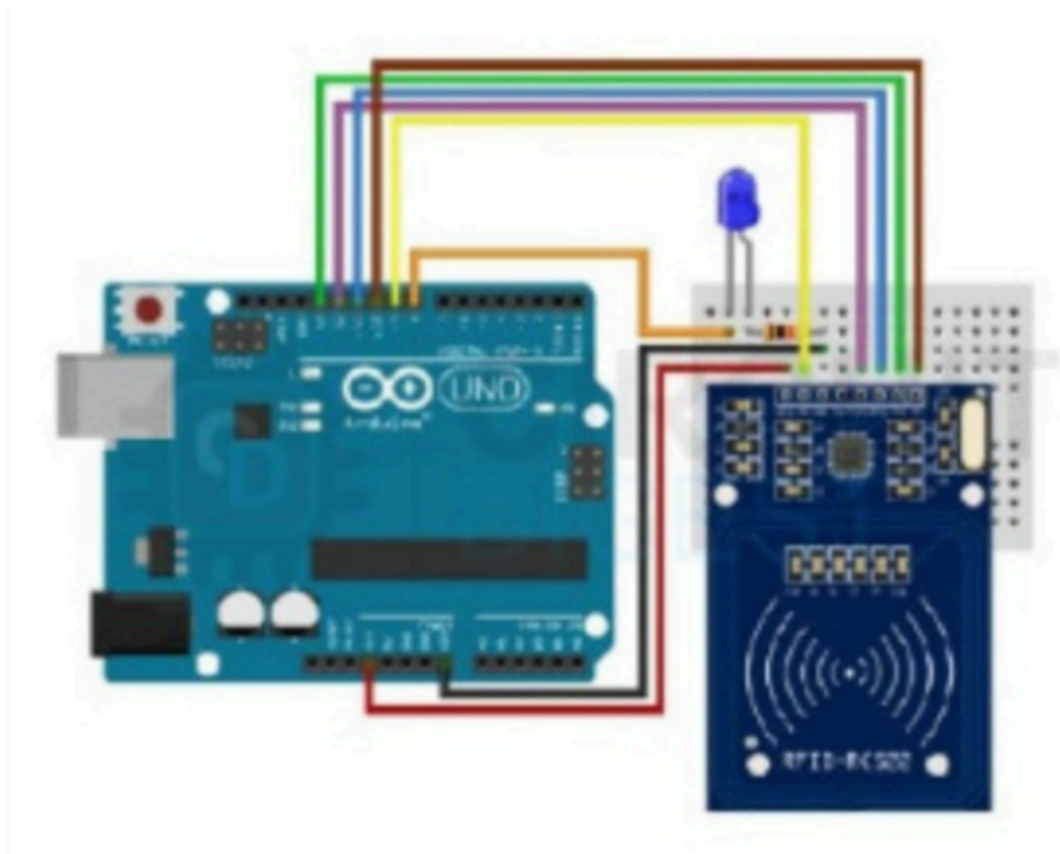
- Guide students through the process of connecting the RFID module and LED circuit, ensuring they understand the basic wiring and component functions (connecting to the "Electric components" stepping stone).
- Facilitate the coding activity, providing support and scaffolding as needed to help students write or modify existing code to interact with the RFID reader and control the LED (connecting to "Basic programming skills" and "Program an Arduino...").
- Encourage students to make observations and record what happens when different RFID tags (if available) are brought near the reader, prompting them to analyze the relationship between the tag and the LED's behavior.
- Facilitate access to the provided video and article about RFID, ensuring students have the necessary tools (devices, internet access, QR code readers).
- Lead discussions after students have engaged with the hands-on activity and the research materials, helping them connect their observations and findings

to the underlying principles of RFID technology and its applications in a smart city.

- Pose probing questions to encourage deeper understanding, such as: "How does the RFID reader know when a tag is present?", "What kind of information might be stored on an RFID tag?", and "How could RFID be used to make a city more sustainable or efficient?"

Expected From Students:

- Follow instructions carefully to connect the RFID module and LED circuit correctly.
- Write or modify code (likely using the Arduino IDE) to read data from the RFID module and control the LED's state.
- Experiment with bringing RFID tags near the reader and carefully observe and record the changes in the LED's behavior.
- Access and engage with the provided video and article to gather information about how RFID technology works and its applications in a smart city.
- Participate actively in discussions, sharing their observations from the circuit activity and insights gained from the video and article.
- Ask questions to clarify their understanding of RFID technology and its potential uses.



Code:

```
#include <SPI.h>
```

```
#include <MFRC522.h>
```

```
// Define RFID module connections (CHECK YOUR WIRING!)
```

```
#define SS_PIN 10
```

```
#define RST_PIN 9
```

```
// Define LED pin (CHECK YOUR WIRING!)
```

```
#define LED_PIN 8
```

```
// Create RFID reader object

MFRC522 rfid(SS_PIN, RST_PIN);


void setup() {

  Serial.begin(9600);

  SPI.begin();

  rfid.PCD_Init();

  pinMode(LED_PIN, OUTPUT);

  Serial.println("Simple RFID LED Blink");

}


void loop() {

  // Check if a new RFID tag is near

  if (rfid.PICC_IsNewCardPresent()) {

    // Check if the tag can be read

    if (rfid.PICC_ReadCardSerial()) {

      Serial.println("Tag Detected!");

      blinkLED(500); // Blink the LED for 500 milliseconds

      rfid.PICC_HaltA(); // Stop reading the current tag

    }

  }

}
```

```

    delay(100); // Wait a bit before checking again
}

// Function to make the LED blink

void blinkLED(int duration) {

    digitalWrite(LED_PIN, HIGH); // Turn LED on

    delay(duration);           // Wait for the specified duration

    digitalWrite(LED_PIN, LOW); // Turn LED off

}

```

Do It! - Teacher Guidance:

- **Preparation:** Ensure you have the necessary components for each student or group: Arduino boards, RFID-RC522 modules, LEDs, resistors, breadboards, and connecting wires. Have example code snippets ready for students to use or modify.
- **Safety:** Emphasize the importance of correct wiring and handling of electronic components.
- **Guidance:** Provide clear, step-by-step instructions for connecting the circuit. Break down the coding task into manageable steps, offering support with syntax and logic. Encourage experimentation within safe parameters.
- **Differentiation:** For students who are struggling, provide pre-written code that they can upload and observe. For more advanced students, challenge them to modify the code to create different LED behaviors based on different RFID tags.

Watch it... - Teacher Guidance:

- **Access:** Ensure students can easily scan the QR code or access the provided URL.
- **Focus:** Before watching, guide students with a question to focus their viewing, such as: "What are some ways RFID technology is being used to improve life in a smart city shown in the video?"
- **Discussion:** After watching, facilitate a discussion to connect the video content to the lesson's focus on smart doors. Ask: "Did you see any

applications of RFID in the video that are similar to how a smart door might work? What other possibilities did you see?"

Read about it... - Teacher Guidance:

- **Access:** Ensure students can easily scan the QR code or access the provided URL.
 - **Reading Strategies:** Encourage students to take notes or highlight key information about how RFID works (e.g., tags, readers, radio waves, data transmission).
 - **Comprehension:** After reading, ask students to explain the basic principles of RFID technology in their own words. Use questions like: "What are the two main components of an RFID system?", "How does the reader get information from the tag?", and "What are some advantages of using RFID?"
-

3. Focus:

Teacher Role:

- Introduce the assessment as an opportunity for students to demonstrate their understanding of RFID technology and its applications, as explored in the lesson.
- Provide clear instructions on how to answer the multiple-choice questions.
- Encourage students to review their notes and the information they gathered during the "Explore" activities if needed.
- Monitor students as they complete the assessment, offering clarification on the questions if necessary but avoiding giving away answers.
- Collect and review student responses to gauge their understanding of the key concepts related to RFID.
- Use the assessment results to identify areas where students have a strong grasp and areas that may require further clarification or reinforcement in future lessons.
- Facilitate a brief discussion after the assessment to review the correct answers and address any misconceptions students might have.

Expected From Students:

- Independently read and analyze each multiple-choice question.

- Apply their understanding of RFID technology, gained from the "Read about it..." activity and the circuit exploration, to select the correct answer for each question.
- Demonstrate their knowledge of the advantages of RFID, its common applications, and the type of RFID tag used with the RC522 module.
- Reflect on their learning throughout the lesson and use the assessment as an opportunity to solidify their understanding.
- Participate respectfully in any post-assessment discussion, asking clarifying questions if needed.

Assessment Questions and Teacher Guidance:

- **Question 1: What is the primary advantage of RFID technology over barcode technology?**
 - A) Lower cost.
 - B) Line-of-sight not required.
 - C) Longer range.
 - D) Higher data storage capacity.
 - **Teacher Guidance:** This question assesses students' understanding of the key benefits of RFID compared to a more familiar technology. Encourage them to recall information from the reading material. The correct answer is **B) Line-of-sight not required.** Discuss why this is a significant advantage in various applications.
- **Question 2: Which of the following is NOT a common application of RFID technology?**
 - A) Inventory tracking.
 - B) Access control.
 - C) Weather forecasting.
 - D) Pet identification.
 - **Teacher Guidance:** This question evaluates students' knowledge of the diverse applications of RFID, likely touched upon in the video and reading. The correct answer is **C) Weather forecasting.** Discuss the applications listed (inventory, access control, pet ID) and why RFID is suitable for them. Briefly explain why RFID isn't typically used for weather forecasting (different types of sensors and data collection methods).
- **Question 3: What type of RFID tag is most commonly used with the RC522 module?**
 - A) Low Frequency (LF).
 - B) High Frequency (HF).
 - C) Ultra High Frequency (UHF).
 - D) Microwave.

- o **Teacher Guidance:** This question tests students' understanding of the specific RFID frequencies and their compatibility with the hardware used in the "Do It!" activity. The RC522 module operates at **B) High Frequency (HF)** (specifically 13.56 MHz). If this wasn't explicitly stated in the student materials, you might need to have provided this information during the "Explore" activity or be prepared to guide students to infer it based on the context of the lesson.
-

4. Practice:

Teacher Role:

- Introduce the servo motor as the component that will simulate the locking/unlocking mechanism of the smart door. Explain that the goal is to make the servo motor move when a valid RFID tag is detected.
- Guide students through the process of connecting the servo motor to the Arduino, referring to the provided circuit diagram. Emphasize the importance of connecting the power, ground, and signal pins correctly.
- Provide students with the necessary Arduino code (or help them modify the previous code) to control the servo motor based on RFID tag detection.
- Encourage students to think about how they would define a "valid" RFID tag (this will likely be explored further in subsequent lessons but can be introduced here conceptually).
- Facilitate experimentation, allowing students to test their circuits and code to see if the servo motor moves correctly when an RFID tag is read.
- Prompt students to consider how this smart door system could be integrated into a larger sustainable city model, as suggested by the "add it to your city" instruction. This can lead to discussions about security, convenience, and potential energy considerations.
- Encourage problem-solving and debugging if students encounter issues with their circuits or code.

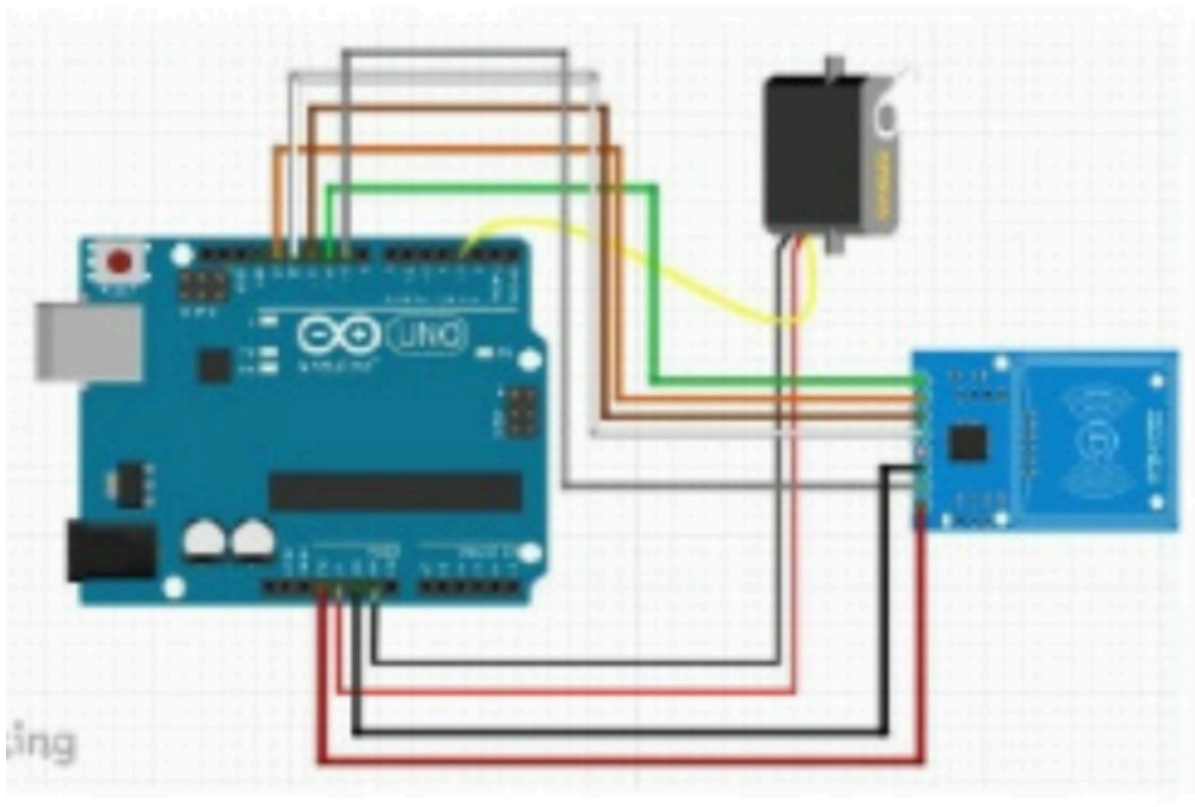
Expected From Students:

- Follow the circuit diagram to correctly connect the servo motor to the Arduino.
- Utilize the provided Arduino code (or modify the previous code) to control the servo motor.
- Observe the movement of the servo motor when an RFID tag is brought near the reader.

- Begin to think about the concept of identifying specific RFID tags for access control.
- Discuss how a smart door system contributes to the functionality and security of a smart and sustainable city.
- Troubleshoot any issues they encounter during the connection and coding process.

Connect the RFID to the Servo Motor - Teacher Guidance:

- **Preparation:** Ensure you have servo motors (SG90 is a common and inexpensive option), jumper wires, and the previously used RFID readers and Arduino boards for each student or group.
- **Circuit Connection:** Guide students to connect the servo motor:
 - Power (usually Red or Orange wire) to Arduino 5V.
 - Ground (usually Brown or Black wire) to Arduino GND.
 - Signal (usually Yellow or White wire) to a digital pin on the Arduino (e.g., Digital Pin 2). **Make sure this pin is correctly defined in the code.**



- **Code Integration:** Provide the following updated Arduino code (or guide students to modify their previous code):

```
#include <SPI.h>

#include <MFRC522.h>

#include <Servo.h> // Include the Servo library


// Define RFID module connections (CHECK YOUR WIRING!)

#define SS_PIN 10

#define RST_PIN 9


// Create RFID reader object

MFRC522 rfid(SS_PIN, RST_PIN);


// Define Servo pin (CHECK YOUR WIRING!)

#define SERVO_PIN 2

Servo myServo; // Create a servo object


void setup() {

  Serial.begin(9600);

  SPI.begin();

  rfid.PCD_Init();

  myServo.attach(SERVO_PIN); // Connect the servo to the pin

  myServo.write(0);          // Set initial servo position (closed)

  Serial.println("Simple RFID Door Control");
```

```
Serial.println("Scan any RFID tag to move the door!");  
  
}  
  
void loop() {  
    // Check if a new RFID tag is near  
    if (rfid.PICC_IsNewCardPresent()) {  
        // Check if the tag can be read  
        if (rfid.PICC_ReadCardSerial()) {  
            Serial.println("Tag Detected - Moving Door!");  
            moveDoor(); // Call the function to move the servo  
            rfid.PICC_HaltA(); // Stop reading the current tag  
        }  
    }  
    delay(100);  
}  
  
// Function to move the servo motor  
void moveDoor() {  
    myServo.write(90); // Move servo to open position (adjust angle as needed)  
    delay(2000);      // Keep door open for 2 seconds  
    myServo.write(0); // Move servo back to closed position  
}
```

Simplified Explanation for Grade 10:

What this code does:

1. Includes Libraries: Like before, these are tools for the Arduino to work with the RFID reader and the servo motor.
2. Defines Pins: We tell the Arduino which pins are connected to the RFID reader and the servo. Make sure these numbers match your wiring!
3. Creates Objects: We create "names" (rfid and myServo) to easily control the RFID reader and the servo motor.
4. setup() Function (Runs Once):
 - o Serial.begin(9600);: Starts the communication to your computer screen.
 - o SPI.begin();: Starts the communication with the RFID reader.
 - o rfid.PCD_Init();: Gets the RFID reader ready.
 - o myServo.attach(SERVO_PIN);: Tells the myServo object which pin the servo is on.
 - o myServo.write(0);: Sets the starting position of the servo (think of this as the door being closed).
 - o Prints a message to the screen.
5. loop() Function (Runs Over and Over):
 - o if (rfid.PICC_IsNewCardPresent());: Checks if an RFID tag is nearby.
 - o if (rfid.PICC_ReadCardSerial());: If a tag is there and can be read...
 - Serial.println("Tag Detected - Moving Door!");: ...it prints a message.
 - moveDoor();: ...and it calls the moveDoor() function to make the servo move.
 - rfid.PICC_HaltA();: Stops reading the tag.
 - o delay(100);: Waits a short time.
6. moveDoor() Function (A Mini-Program for the Servo):
 - o void moveDoor(): This is a set of instructions to move the servo.
 - o myServo.write(90);: This tells the servo to move to a certain angle (think of this as opening the door). You might need to change 90 to a different number depending on how your servo is connected.
 - o delay(2000);: It waits for 2 seconds (the "door" stays open).
 - o myServo.write(0);: It tells the servo to move back to the starting position (closing the "door"). Again, 0 might need adjustment.

How to Use:

1. Install Library: Make sure the Servo and MFRC522 libraries are installed.

2. **Connect Wires:** Connect the RFID reader and the servo to the Arduino, being careful with the power, ground, and signal wires. Double-check your pin numbers match the code!
3. **Upload Code:** Copy this code to the Arduino software and upload it.
4. **Open Serial Monitor:** Open the Serial Monitor to see messages.
5. **Test:** Bring *any* RFID tag close to the reader. You should see the message on the screen, and the servo motor should move back and forth, simulating the door opening and closing.

Objectives:

- 1-Know how mobile apps control things wirelessly.
- 2-Understand basic app design for control.
- 3-Learn how apps connect to devices via Bluetooth.
- 4-Program a simple app to send commands.
- 5-Connect an app to Arduino to control a device.

☐ Discuss how app control makes a city "smart."☐ Discuss RFID uses in smart cities.

Challenge Questions of the Lesson

How can we use a phone app to simply control something in a "smart city"?

SEL relation

Self-Awareness: Think about how tech can make things easier and how you felt creating the app.

Self-Management: Focus on building the app step-by-step and solving problems.

Social Awareness: Consider how remote control can help people in a smart city.

Relationship Skills: Share app ideas and help others with their code.

Responsible Decision-Making: Think about the benefits and risks of controlling things with apps.

Here are short and simple answers:

By creating an app that sends wireless signals (like Bluetooth) to a device, telling it what to do (on/off, etc.).

1. Engage:

Teacher Role:

- Initiate a discussion about the concept of a "smart home" and elicit students' prior knowledge of smart devices they may have encountered or heard about (e.g., smart lights, thermostats, security systems). Connect this to the broader theme of a "smart city."
- Incorporate engaging questions like: "What are some tasks you wish you could control remotely using your phone?", "What are the benefits of having a smart home?", and "What are some potential security concerns associated with smart home devices?"
- Show visuals of various smart home devices and interfaces on mobile apps that control them.
- Pose the "Let's Think" question to the class and encourage initial brainstorming and predictions about the features and functionalities such an app might have.
- Briefly introduce the idea of using a mobile application to interact with physical devices, hinting at the role of programming and communication technologies like Bluetooth.

Expected From Students:

- Actively share their understanding and experiences with smart home technology.
- Respond to open-ended questions with thoughtful ideas and examples.
- Engage with the visuals of smart home devices and control apps.
- Begin to think critically about the "Let's Think" question and offer initial ideas for a mobile app to control a home.
- Express curiosity about how a mobile app can interact with physical devices.

Preview with a Hook:

- **Strategy:** Begin with a demonstration or a compelling scenario.
- **Implementation:**
 - **Scenario:** Describe a situation where someone forgets to turn off a light at home and then uses their phone to do it remotely.
 - **Visual:** Show a short video clip of a person controlling a smart light or another simple smart home device using a mobile app.
 - Ask: "How do you think the phone is able to communicate with the light in this way?"

Strategies to use with this part:

- **Brainstorming:** Have students brainstorm a list of devices in a home that could be made "smart" and controlled remotely.
 - **Think-Pair-Share:** Pose a question like, "What are the advantages of controlling your home remotely?", have students think individually, discuss with a partner, and then share with the class.
 - **Real-World Connections:** Ask students if they know anyone who uses smart home devices and have them share their experiences (if any).
 - **Concept Mapping:** Start a concept map on the board with "Smart City" and branch out to "Smart Home" and then potential controllable devices.
-

2. Explore:

Teacher Role:

- Introduce the concept of user interface (UI) design for mobile applications. Explain that the interface is what the user sees and interacts with to control the smart home devices.
- Guide students to analyze the provided sample app interface, discussing the purpose of each button and element.
- Facilitate a brainstorming session where students think about the specific smart home devices they want to control with their app and the necessary controls for each device. Encourage them to consider user-friendliness and intuitive design.
- Provide materials for students to sketch out their own mobile app interfaces. This could include paper and pencils, graphic design software if available, or even simple prototyping tools.
- Ensure students have access to the video resource on creating smart home mobile applications. Encourage them to take notes on key design considerations and functionalities discussed in the video.
- After the design phase and video viewing, lead a discussion where students share their app interface designs and explain their choices. Encourage peer feedback and constructive criticism based on usability and functionality.

Expected From Students:

- Understand the importance of UI design in creating user-friendly mobile applications.
- Analyze the provided sample app interface and identify its basic functionalities.

- Brainstorm and decide on the smart home devices they want to control with their app.
- Design a visual interface for their mobile application, including buttons, labels, and layout.
- Access and engage with the provided video resource on creating smart home mobile applications.
- Share their app interface designs with the class and provide/receive feedback on their designs.

Do It! - Teacher Guidance:

Follow this video step by step:

<https://www.youtube.com/watch?v=IOUwl4TIKLo>

- **Emphasis on Planning:** Stress that this stage is about planning the user experience before actual coding.
- **Device Variety:** Encourage students to think beyond the examples (lights, fan) and consider other smart devices relevant to a home (e.g., thermostat, door lock, speaker).
- **Sketching First:** Recommend starting with paper sketches to quickly iterate on ideas before moving to digital tools (if available).
- **Usability:** Guide students to think about clear labeling, button size, and logical flow for controlling devices.

Watch it... - Teacher Guidance:

- **Pre-Viewing Questions:** Before students watch the video, pose questions to focus their viewing, such as: "What are some key elements to consider when designing a smart home app?", "What functionalities are commonly included in such apps?", and "What tools or platforms might be used to create these apps?" (This can subtly hint at App Inventor if it's the intended platform).
- **Note-Taking:** Encourage students to note down design principles, common features, and any specific tools or methods mentioned in the video.
- **Post-Viewing Discussion:** After watching, discuss the video's content and how it relates to their own app designs. Ask: "What ideas from the video did you incorporate into your design?", "What were some important design considerations highlighted in the video?", and "Did the video give you any new ideas for your app?"

3. Assessment:

Teacher Role:

- Introduce the assessment as a way for students to check their understanding of smart home mobile applications and the technologies involved in controlling smart devices.
- Provide clear instructions on how to choose the correct answer for each multiple-choice question.
- Encourage students to think back to the "Engage" discussion and the information they gathered during the "Explore" activities (especially the video) to answer the questions.
- Monitor students as they complete the assessment, offering clarification on the questions if needed but avoiding giving away answers.
- Review student responses to identify their understanding of key concepts and any areas where they might need further clarification.
- Facilitate a brief discussion after the assessment to go over the correct answers and address any misconceptions.

Expected From Students:

- Read and analyze each multiple-choice question carefully.
- Apply their understanding of smart home mobile applications and related technologies to select the best answer.
- Demonstrate their knowledge of common features of smart home apps and the communication technologies used.
- Reflect on what they have learned in the lesson.

Focus - Assessment Questions and Teacher Guidance:

- **Question 1: What is a common feature of smart home mobile applications?**
 - A) Ability to control multiple devices from different manufacturers.
 - B) Requirement of physical buttons for operation.
 - C) Limited functionality to one device only.
 - D) No remote access capabilities.
 - **Correct Answer: A) Ability to control multiple devices from different manufacturers.**
 - **Teacher Guidance:** This question assesses students' understanding of the core convenience and functionality of smart home hubs and applications. Discuss how a central app can integrate various devices.

Options B, C, and D are generally contrary to the nature of smart home systems.

- **Question 2: Which of the following technologies is commonly used to control smart home devices via a mobile application?**
 - A) Ethernet cables.
 - B) Bluetooth and Wi-Fi.
 - C) USB connections.
 - D) Analog signals.
 - **Correct Answer: B) Bluetooth and Wi-Fi.**
 - **Teacher Guidance:** This question focuses on the communication methods between a mobile app and smart home devices. Emphasize the wireless nature of most smart home communication. Discuss the roles of Bluetooth (for shorter-range, direct connections) and Wi-Fi (for wider network connectivity and internet access). Explain why Ethernet and USB are less common for direct mobile app control of *in-home* devices. Analog signals are not typically used for digital control in this context.
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4. Practice:

Teacher Role:

- Introduce MIT App Inventor as the platform students will use to create their mobile application. Provide guidance on accessing and navigating the App Inventor interface.
- Explain the concept of block-based programming in App Inventor and how it translates user interface actions (like button clicks) into code.
- Guide students to create the basic UI they designed in the "Explore" section within App Inventor, adding buttons for controlling their chosen smart home devices.
- Introduce the Bluetooth client component in App Inventor and explain its role in establishing communication with the Arduino.
- Provide students with example App Inventor blocks (similar to the visual on the page) that demonstrate:
 - Connecting to a Bluetooth device (the Arduino).
 - Sending data (text commands like "1" and "0") when buttons are clicked.
 - (Optional, for more advanced students) Receiving data back from the Arduino (if the Arduino is sending status updates).

- Provide the corresponding Arduino code that listens for Bluetooth commands and controls the connected "smart home device" (in our example, an LED).
- Facilitate the process of connecting the App Inventor app (via Bluetooth) to the Arduino and testing the remote control functionality.
- Encourage students to troubleshoot any issues they encounter during the coding and connection process.
- Optionally, challenge students to add more features or controls to their app.

Expected From Students:

- Use MIT App Inventor to create the user interface for their smart home control application.
- Understand and utilize the Bluetooth client component in App Inventor.
- Write block code in App Inventor to send commands to the Arduino based on button clicks.
- (Optional) Write block code to receive data from the Arduino.
- Connect their App Inventor app to the Arduino via Bluetooth.
- Observe and verify the remote control of the connected "smart home device."
- Troubleshoot any coding or connectivity problems.

Write the code for your application and connect it to Arduino - Teacher Guidance:

- **App Inventor UI Design:** Guide students to create buttons in the App Inventor Designer view that correspond to the controls for their chosen devices (e.g., "Light ON," "Light OFF," "Fan Speed Up," "Fan Speed Down").
- **App Inventor Block Coding (Example for LED Control):**
 - **Bluetooth Connection:** Show how to use the `BluetoothClient.Connect` block (likely triggered by the "Click here" button in the sample UI) to connect to the Arduino's Bluetooth module.
 - **Sending Commands (for "Light ON" button):**
 - When `Button_LightOn.Click`
 - do `BluetoothClient.SendText` with the text value "1".
 - **Sending Commands (for "Light OFF" button):**
 - When `Button_LightOff.Click`

- do `BluetoothClient.SendText` with the text value "0".

● **Arduino Code (Example for LED Control via Bluetooth):**

Arduino

```
#include <SoftwareSerial.h>

// Define Bluetooth module pins (adjust if needed)
#define RX_PIN 10
#define TX_PIN 11
SoftwareSerial bluetoothSerial(RX_PIN, TX_PIN);

// Define LED pin (adjust if needed)
#define LED_PIN 8

void setup() {
  Serial.begin(9600);
  bluetoothSerial.begin(9600);
  pinMode(LED_PIN, OUTPUT);
  digitalWrite(LED_PIN, LOW); // Initially off
  Serial.println("Bluetooth LED Control Ready");
}

void loop() {
  if (bluetoothSerial.available()) {
    char command = bluetoothSerial.read();
    Serial.print("Received command: ");
    Serial.println(command);
    if (command == '1') {
      digitalWrite(LED_PIN, HIGH); // Turn LED ON
    } else if (command == '0') {
      digitalWrite(LED_PIN, LOW); // Turn LED OFF
    }
  }
}
```

- **Bluetooth Setup:** Ensure students understand that they will need a Bluetooth module (like HC-05 or HC-06) connected to their Arduino. Guide them on the basic wiring of the Bluetooth module (VCC, GND, RX to Arduino TX, TX to Arduino RX).
- **Connecting App to Arduino:** Provide instructions on how to pair their Android devices with the Arduino's Bluetooth module and how to select the connected Bluetooth device in the App Inventor app.
- **Troubleshooting:** Be prepared to assist students with common issues like Bluetooth connection problems, incorrect App Inventor block logic, or errors in the Arduino code.

